



MONASH University

Melt, Crevassing, Buttressing: Assessing the Cook Ice Shelf's Vulnerability to Climate Change

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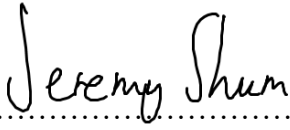
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Declaration

This thesis/report contains no material which has been accepted for the award of any other degree or diploma in any university and that, to the best of the candidate's knowledge and belief, the thesis contains no material previously published or written by another person, except when due reference is made in the text.

Signature: 

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Chapter 1 - Abstract

Antarctic ice shelves play a crucial role in regulating global sea levels by exerting buttressing stresses that restrain the outflow of grounded ice into the ocean. Ice shelf thinning by ocean-driven basal melting or calving by crevasse and rift propagation reduce this resistive buttressing force, leading to enhanced ice discharge and sea level rise. Ocean-driven basal melting is one of the dominant processes driving Antarctic ice loss, yet remains poorly constrained due to sparse in-situ measurements and uncertainties in remote-sensing and model-derived melt estimates. Furthermore, the occurrence and stress drivers of crevasses remain poorly constrained. Additionally, the buttressing capacity of Antarctic shelves has been quantified at coarse resolution only, limiting understanding of fine-scale variability. In particular, the Cook Ice Shelf – the major outlet of the Wilkes Subglacial Basin (WSB; ~ 3.02 m sea level equivalent) – is among the least studied regions of East Antarctica. Its melt regime, crevassing behaviour, and buttressing dynamics are poorly constrained, despite its vulnerability to climate warming.

This thesis aims to assess the present-day vulnerability of the Cook Ice Shelf to climate change, through assessing how its buttressing evolves in response to ocean-driven melt and crevassing. This aim is addressed through three research questions: (1) what is the magnitude and mode of ocean-driven basal melting at the Cook Ice Shelf; (2) what regions are vulnerable to crevasse formation and subsequent calving; and (3) how has the Cook Ice Shelf flow regime evolved in response to ocean-driven melting and crevassing over recent decades, and what do these changes reveal about the buttressing capacity of this shelf?

I find that the Cook Ice Shelf has low basal melt rates primarily driven by modified Circumpolar Deep Water acting near the grounding line. The eastern shear margin, grounding line, and far west of the Cook Ice Shelf are most vulnerable to crevasse formation. The buttressing capacity

and passive ice boundary changed little between 2006-2018. This suggests that the Cook Ice Shelf has not yet exhibited a significant response to present-day climate forcing.

However, basal melt rate estimates are highly uncertain due to coarse satellite-derived products near the grounding line, the absence of direct in-situ measurements, and poorly constrained cavity bathymetry in ocean models. Given these uncertainties in basal melt rates, and combined with the findings from the three research questions, a decision-making framework was developed to identify optimal locations for Autonomous phase-sensitive Radio Echo Sounder (ApRES) deployment - an instrument that measures in-situ melt rates. The framework combines basal melt rates, crevassing, buttressing, and surface mass balance to define a weighted suitability score, which identifies the southern-central region of the Cook Ice Shelf as the most ideal deployment site. This study provides an approach guiding in-situ instrument deployment across Antarctic ice shelves, which can be readily applied to other variables and locations to help guide efforts for data collection, such as the *Antarctica InSync* initiative for circumpolar measurements.

Keywords: Cook Ice Shelf, ocean-driven basal melting, crevassing, calving, and buttressing

Chapter 2 - Introduction

Antarctica holds 60% of the global total freshwater (BAS, 2025), and its stability is crucial for regulating global sea levels. As anthropogenic warming progresses, the ocean-driven basal melting of ice shelves increases, driving Antarctic ice loss and consequent sea level rise. However, ice shelves play an essential role in restraining the flow of grounded ice into the ocean. Therefore, understanding how dynamic changes to ice shelves affect their vulnerability to climate change is paramount for assessing future sea level rise.

Floating ice shelves play a critical role in providing stabilising forces - known as buttressing - on the upstream grounded ice sheet (Gudmundsson et al., 2019) (figure 1), reducing the outflow from the grounded ice sheet into the ocean. As ice shelves melt due to ocean-induced thinning, the buttressing stresses can reduce, resulting in increased ice discharge at the ice-ocean interface and sea level rise (Gudmundsson et al., 2019). Ice shelf calving can also result in an instantaneous reduction in buttressing (Greene et al., 2022), with increases in flow speeds that can persist for years to decades after the event (Scambos et al., 2004). However, not all regions of an ice shelf play an active role in dynamically buttressing the upstream ice - ice shelf regions that do not provide buttressing and can therefore be removed without causing major dynamic implications are referred to as passive ice regions (Fürst et al., 2016). The passive ice boundary separates regions of active buttressing from passive ice. Because ice shelves control the flow of grounded ice into the ocean, changes in their thickness and extent can have implications on the buttressing of active ice, influencing ice sheet stability and global sea level rise.

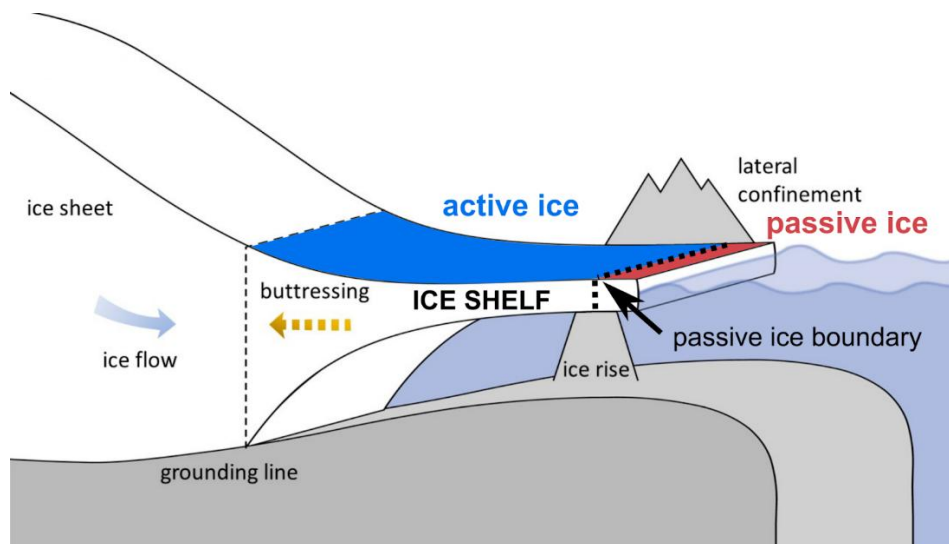


Figure 1: The grounded ice sheet transitions to a floating ice shelf at the grounding line - the point where the ice detaches from the bed and begins to float in the ocean. Buttressing forces that stabilise the upstream grounded ice sheet are generated due to the interaction between the ice shelf and topography, or lateral confinement in embayed ice shelves (modified from (Reese, 2019)).

The region of interest for this thesis is the Cook Ice Shelf (figure 2). The Cook Ice Shelf is situated downstream of the Cook Glacier system in East Antarctica, and acts as a fundamental drain to the Wilkes Subglacial Basin (WSB) (Lee et al., 2023). The Cook Ice Shelf is the main

buttressing ice shelf of the WSB. The WSB is one of the largest marine basins in East Antarctica, and holds the equivalent of 3.02 metres of global sea level rise (Rignot et al., 2019). Paleoclimate records show that the WSB has undergone repeated retreat events - around 330,000 and 125,000 years ago - linked to warmer interglacial Southern Ocean temperatures and higher global sea levels (Crotti et al., 2022), highlighting that the WSB is one of the primary drivers of East Antarctic ice loss under warming climates. Given the Cook Ice Shelf's central role in buttressing the upstream WSB and its potential influence on future sea level rise, understanding the controls on its stability and vulnerability to collapse in a warming climate is critical. Despite its importance in regulating ice outflow from the Cook Glacier and therefore the WSB, the Cook Ice Shelf remains poorly constrained, with limited understanding of its melt dynamics, vulnerability to crevassing, and buttressing capacity.

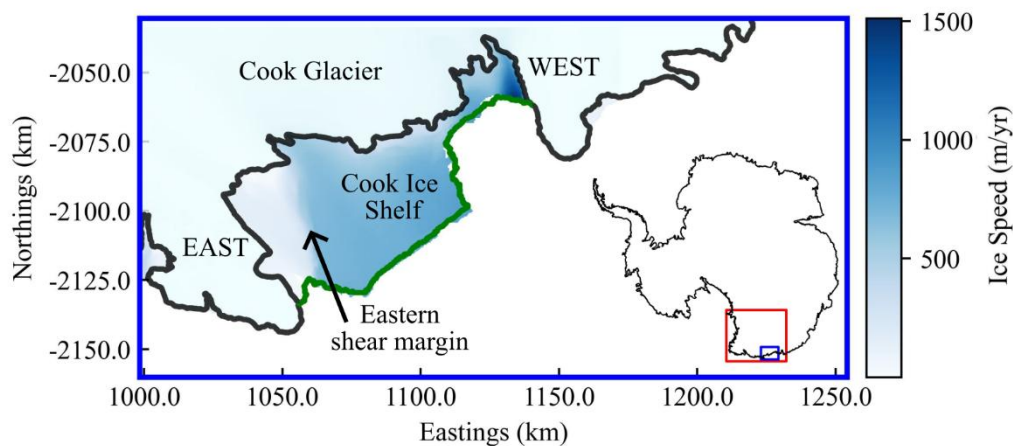


Figure 2: The spatial context of the WSB (red box) and the Cook Ice Shelf (blue box) in Antarctica. The ice velocity of the Cook Ice Shelf is presented (Rignot et al., 2017). The 2003 grounding line is plotted as a solid black line (Bindschadler and Choi, 2011) and the 2006 ice front is plotted as a solid green line (Greene et al., 2024).

Although ocean-driven basal melting is one of the dominating controls on the present-day vulnerability of the Cook Ice Shelf, its magnitude, spatial variability, and underlying mechanisms remain poorly understood. Existing studies disagree on both the overall magnitude and variability in ocean-driven melt of the Cook Ice Shelf. A recent study suggests that the Cook Ice Shelf has thickened overall, due to a reported net mass gain of 470 Gt from 1997 to 2021, despite an observed mean basal melt rate of 1.18 m/yr from 2010 to 2021 (Davison et al., 2023).

However, this contrasts with previous work which suggests that the Cook Ice Shelf has experienced highly heterogeneous melt-induced thinning patterns, where the western region of the Cook Ice Shelf has undergone upstream thinning since 2010, yet lacks basal melt observations in the eastern region of the Cook Ice Shelf (Miles et al., 2018). Importantly, these studies were based on satellite-derived estimates of melt, which can be inaccurate in regions close to the grounding line or where the ice shelf isn't in hydrostatic equilibrium (Bird et al., 2024). Basal melting of ice shelves occurs when warmer ocean water circulates into sub-ice-shelf cavities, driven by changes in the properties and incursions of different water masses from changing wind stress patterns. There are three main water parcel types (modes of melt) found in the Southern Ocean; Mode 1 refers to Dense Shelf Water (DSW) which is produced from sea ice in coastal polynyas, Mode 2 is defined as Circumpolar Deep Water (CDW) transformed into modified Circumpolar Deep Water (mCDW), and Mode 3 relates to fresher Antarctic Surface Water (ASW) (McCormack et al., 2024). Ocean-driven basal melt is predominantly caused by the warmer Mode 2, where changing wind stress patterns under climate change can transport CDW onto the continental shelf (where it mixes with shelf water and becomes mCDW) (McCormack et al., 2024), and into ice shelf basal cavities. The presence of mCDW at the Cook Ice Shelf is suspected (Constantino and Tinto, 2023), however this has yet to be spatially constrained and linked to basal melt rate. The last oceanographic in-situ data from the Cook Ice Shelf dates back to 1957-58 (Constantino and Tinto, 2023), where no mCDW was observed at the time. However, other studies suggest that mCDW is highly variable and critically linked to wind stress variability (Greene et al., 2017). There has been a lack of ship-based conductivity-temperature-depth surveys since the 1957-58 period (Constantino and Tinto, 2023), creating uncertainties in melt rates due to this sparsity of observations. Collectively, these uncertainties highlight a critical knowledge gap in quantifying both the magnitude and mode of ocean-driven basal melting beneath the Cook Ice Shelf.

Crevassing and calving are key processes that influence the structural integrity and buttressing capacity of the Cook Ice Shelf, yet their occurrence, spatial variability, and drivers remain poorly constrained. Crevasses are brittle fractures that form from stress within an ice body (Colgan et al., 2016), and can pre-condition ice shelves and act as predictors for hydrofracture and subsequent calving events (Surawy-Stepney et al., 2023). Observations at the Cook Ice Shelf between 1947 and 2017 indicate minimal crevassing in the eastern sector, but extensive and persistent crevassing across the western region (Miles et al., 2018). This spatial variability is reflected in calving patterns: while no major calving events have occurred near the eastern Cook Ice Shelf since 1973, the western sector experienced a near-complete collapse between 1973 and 1989 (Miles et al., 2018). Losses in buttressing can also result from calving events through the propagation of crevasses (Greene et al., 2022). However, crevasse fields across Antarctica, including the Cook Ice Shelf, remain poorly mapped (Zhao et al., 2022), and factors that precondition shelf vulnerability to crevassing such as strain thinning (Vaughan, 1993), are poorly constrained. No efforts have been made to assess the probability of crevasse formation and expansion throughout the Cook Ice Shelf, representing a significant knowledge gap in predicting future calving and ice shelf instability.

Uncertainties in ocean-driven basal melting and calving at the Cook Ice Shelf restrict our capacity to assess the role of the Cook Ice Shelf in buttressing the upstream WSB. The buttressing capacity of the Cook Ice Shelf has been quantified (Fürst et al., 2016), but at coarse resolution. This highlights a lack of knowledge of the fine-scale spatial and temporal variability in buttressing across the Cook Ice Shelf and how this impacts the discharge from the upstream catchment across the grounding line. High-resolution velocity datasets are available, allowing for detailed quantification of ice flux across the grounding line and the resulting buttressing forces. However, these datasets are yet to be used to assess the buttressing capacity of the Cook

Ice Shelf, posing a critical knowledge gap in understanding how melt and calving processes at this shelf influence ice discharge from the WSB.

Overall, the drivers and magnitude of melt, crevassing, and buttressing of the Cook Ice Shelf are insufficiently quantified. These knowledge gaps limit the ability to assess the vulnerability of the Cook Ice Shelf - WSB system to climate change forcings, and future contribution to sea level rise. The limited understanding of melt, crevassing, and buttressing across the Cook Ice Shelf underscores the need for targeted in-situ observations, which are essential to both constrain basal melt rates and to improve process-based assessments of ice shelf vulnerability under a warming climate, as well as to inform the design and feasibility of field campaigns.

The overarching aim of this thesis is to assess the present-day vulnerability of the Cook Ice Shelf to climate change, through reduced buttressing in response to ocean-driven melt and crevassing. I address this aim through answering three core research questions:

1. What is the magnitude and mode of ocean-driven basal melting at the Cook Ice Shelf?
2. What regions of the Cook Ice Shelf are vulnerable to crevasse formation and subsequent calving events?
3. How has the Cook Ice Shelf flow regime evolved in response to ocean-driven melting and crevassing over recent decades, and what do these changes reveal about the buttressing capacity of this shelf?

Given the fundamental importance of in-situ measurements in improving the process-based understanding of the Cook Ice Shelf - WSB system, I use the insights gained from the three research questions to develop a decision-making framework for guiding targeted instrument deployment. This approach is motivated by opportunities offered through the Antarctica InSync program, a coordinated international initiative conducting circumpolar measurements across Antarctica from 2027 to 2029.

Chapter 3 - Methods

3.1 Melt Regime

To address the first research question, I investigate the magnitude of basal melting at the Cook Ice Shelf to determine whether it is contributing to ice shelf thinning, and classify the dominant modes of melt. To do this, I use ocean temperature and salinity, and ice shelf basal melt rate estimates taken from the Whole Antarctic Ocean Model (WAOM) (Richter et al., 2022). WAOM uses the Regional Ocean Modeling System (ROMS) (Shchepetkin and McWilliams, 2005), an open-source ocean model. WAOM is the highest spatial resolution ocean model output (2 km) currently available for Antarctica, and resolves ice shelf cavity dynamics. The output used in this study covers the approximate domain from 1,032,750 to 1,136,318 E (152.30 to 153.63° longitude) and -2,190,415 to -2,058,869 N (-68.83 to -67.77° latitude). Bathymetry and cavity geometry are derived from Bedmap2, which incorporates the bathymetry from the 2008 GEBCO grid (Fretwell et al., 2013). The WAOM has a slightly warm and salty bias (Richter et al., 2022), and holds notable uncertainties in both bathymetry (88 to 320 m) and ice thickness (150 to 200 m) (Fretwell et al., 2013). These uncertainties are derived from continent-wide Bedmap2 data and therefore do not specifically represent conditions beneath the Cook Ice Shelf. The model employs 31 terrain-following vertical layers with higher resolution near the surface and the bed, ranging from 30 to 50 m (Richter et al., 2022). Surface forcing is applied using daily heat and salt fluxes derived from observed sea ice concentration and daily wind stress calculated from ERA_Interim (Richter et al., 2022). Surface temperature and salinity are relaxed towards the Southern Ocean State Estimate (SOSE), while open boundary conditions are derived from ECCO2 reanalysis, providing monthly sea surface height, temperature, salinity, and velocity fields (Richter et al., 2022). Further model configuration details are described in (Richter et al., 2022).

The WAOM outputs analysed here are potential temperature ($^{\circ}\text{C}$) and practical salinity (PSU), with their corresponding depths (converted into sea pressure), latitude, and longitude locations. To classify the water parcels across the domain of interest, I first calculate neutral density using the GSW python package from TEOS-10 (McDougall and Barker, 2011, Jackett and McDougall, 1997), where the GSW algorithm converts potential temperature to in-situ temperature, and practical salinity to absolute salinity. This calculated output of neutral density, in addition to potential temperature and practical salinity, are constraints required to define the type of water parcels present at the Cook Ice Shelf and surrounding region (table 1). Mode 1 melting by DSW is characterised by the potential temperature range of -1.9°C and -1.8°C , a salinity value higher than 34.5 PSU, and a neutral density value greater than 28.27 kg m^{-3} (Narayanan et al., 2019). Mode 2 melting by mCDW is classified by potential temperatures between -1.8 and 0°C , and a neutral density range between 28 kg m^{-3} and 28.27 kg m^{-3} (Narayanan et al., 2019). Mode 3 melting by ASW is defined by neutral densities less than 28 kg m^{-3} (Ribeiro et al., 2023). The definitions for DSW and mCDW are extracted from literature relating to Vincennes Bay (Narayanan et al., 2019), and ASW relating to the Shackleton-Denman System (Ribeiro et al., 2023). These regions are in East Antarctica but not relate to the Cook Ice Shelf specifically. However, these water parcel definition ranges are widely accepted and align with literature.

Table 1: Definition of water masses according to their potential temperature ($^{\circ}\text{C}$), practical salinity (PSU), and neutral density (kg m^{-3}).

Water Mass	Potential Temperature ($^{\circ}\text{C}$)	Practical Salinity (PSU)	Neutral Density (kg m^{-3})	Reference
DSW	$-1.9 < T < -1.8$	$S > 34.5$	$\text{ND} > 28.27$	(Narayanan et al., 2019)
mCDW	$-1.8 < T < 0$	-	$28 < \text{ND} < 28.27$	(Narayanan et al., 2019)
ASW	-	-	$\text{ND} < 28$	(Ribeiro et al., 2023)

To investigate the modes driving melting at the Cook Ice Shelf, I first plot the basal melt rates spatially across the Cook Ice Shelf (figure 3). From there, I identify zones of higher melt rate, and extract temperature, salinity, and neutral density data in three cross sections extending ~ 120

km seaward from the grounding line (denoted A-A', B-B', and C-C' cross sections, see table 2). Line A in particular cuts through a basal melt rate maximum near the grounding line and an inner shelf body of lower basal melt rates, providing insight into regions with varying basal melt rate levels. I then use the potential temperature, practical salinity and neutral density data to classify the water masses along these cross sections, using these to define the melt modes.

Table 2: The starting and ending coordinates for Lines A-A', B-B', and C-C' in Eastings (m) and Northings (m), where X refers to the starting point, and X' refers to the ending point.

Line	Start – Eastings (m)	Start – Northings (m)	End – Eastings (m)	End – Northings (m)
A	1078998.19	-2072930.45	1134587.87	-2161360.53
B	1049863.97 E	-2074869.56	1112546.39	-2174544.83
C	1034534.06	-2086972.74	1096302.03	-2185202.45

3.2 Crevassing

To address the second research question and determine the vulnerability of the Cook Ice Shelf to crevassing and subsequent calving, I need to determine the location of current crevasses and investigate the strain history of the ice shelf, which relates to fracture formation. To achieve this, I use satellite imagery to derive surface crevasses and surface velocities that can be used to calculate effective strain rates. Specifically, yearly mosaics of the MEaSUREs velocities (Mouginot et al., 2017) are used to calculate strain rates. I use yearly data from 2006 to 2018 as it represents the more recent decade of data available. The grounding line is derived from the US Antarctic Program Data Centre (USAP-DC) (Bindshadler and Choi, 2011), and is kept fixed for each year, and yearly MEaSUREs ITS_LIVE ice fronts (Greene et al., 2024) are used. I make use of the input functions from the Ice-sheet and Sea-level System Model (ISSM; (Larour et al., 2012)) to calculate $\dot{\epsilon}_{xx}$ (longitudinal normal strain rate in the x-direction), $\dot{\epsilon}_{yy}$ (longitudinal normal strain rate in the y-direction), and $\dot{\epsilon}_{xy}$ (shear strain rate in the xy plane) for the Cook Ice Shelf for each year of data, using a Thomas analytical equation (Thomas, 1973), a smoothing parameter of 0, and a longitudinal coordinate system which was aligned along a flowline at every point. From this, the effective strain rate ($\dot{\epsilon}_{eff}$) could be calculated for each year using the equation: (Larour et al., 2012)

$$\dot{\epsilon}_{eff} = \sqrt{\frac{1}{2}(\dot{\epsilon}_{exx}^2 + \dot{\epsilon}_{eyy}^2) + \dot{\epsilon}_{exy}^2} \quad (1)$$

The median $\dot{\epsilon}_{eff}$ was plotted and used in following calculations to ensure experimental robustness.

To define regions of crevassing, I used the Landsat Image Mosaic of Antarctica (LIMA), which was constructed from 1999-2003 Landsat-7 ETM+ imagery at 15 m resolution (Bindschadler et al., 2008). Surface crevasses were visually identified and traced in QGIS by scanning the LIMA imagery for characteristic linear and curvilinear fracture patterns, distinguishing these crevasses from shadows and other surface features based on contrast and continuity. LIMA was the chosen satellite imagery used, as it provided the clearest image at the highest resolution available. I was unable to access more recent high-resolution satellite imagery (e.g. Sentinel).

In python, kernel density estimations of $\dot{\epsilon}_{eff}$ were calculated for both the crevassed regions and the entire Cook Ice Shelf, and then normalised to produce probability density functions (PDFs), representing the distribution of probabilities across the range of $\dot{\epsilon}_{eff}$ values. The median PDFs were computed separately for the crevassed regions and for the entire shelf. By separating the Cook Ice Shelf into these two categories, the conditional probability of crevasse formation given an $\dot{\epsilon}_{eff}$ value can be calculated, according to the conditional probability equation of:

$$P(\text{crevasse} | \dot{\epsilon}_{eff}) = \frac{P(\dot{\epsilon}_{eff} | \text{crevasse})}{P(\dot{\epsilon}_{eff})} P(\text{crevasse}) \quad (2)$$

where $P(\dot{\epsilon}_{eff} | \text{crevasse})$ is the conditional probability of observing a given $\dot{\epsilon}_{eff}$ in a crevassed region, $P(\dot{\epsilon}_{eff})$ is the probability of that $\dot{\epsilon}_{eff}$ value across the Cook Ice Shelf, and $P(\text{crevasse})$ is the probability of a certain region being crevassed already.

3.3 Buttrressing

To address the third research question by understanding how ocean-driven melting and crevassing influence the buttressing capacity of the Cook Ice Shelf, I need to determine regions of active and passive ice, and the ice outflux of the shelf. To achieve this the maximum buttressing number is calculated, which represents the maximum value of the buttressing number along any principal axis - it is not direction dependent. This is defined in (Fürst et al., 2016) and (Larour et al., 2012) as follows:

$$N_0 = \frac{1}{2} \rho_i \left(1 - \frac{\rho_i}{\rho_w} \right) gH \quad (3)$$

$$N(\max) = [(2\tau_{xx} + \tau_{yy})n_x + \tau_{xy}n_x]n_x + [\tau_{xy}n_x + (\tau_{xx} + 2\tau_{yy})n_y]n_y \quad (4)$$

$$knmax = 1 - \frac{N(\max)}{N_0} \quad (5)$$

where N_0 is the ice shelf-free back-stress value (Gudmundsson, 2013), ρ_i is the density of ice (917 kg/m³), ρ_w is the density of ocean water (1023 kg/m³), g is the acceleration due to gravity (9.81 m/s²), and H is the ice thickness (m). $N(\max)$ represents the resistive normal stress of the ice shelf itself, n_i represent the magnitude of stress along the secondary principal axis, and τ_{ij} is the deviatoric stress tensors (Gudmundsson, 2013). The maximum buttressing number Eqs. (5) is then calculated as $knmax$ by the fraction of Eqs. (3) over Eqs. (4). To calculate these quantities, I use the 2003 grounding line from USAP-DC (Bindschadler and Choi, 2011), annual MEaSUREs ITS_LIVE ice fronts (Greene et al., 2024), annual MEaSUREs velocity maps (Mouginot et al., 2017), annual MEaSUREs ITS_LIVE ice thicknesses (Paolo et al., 2024), and surface elevations from ERS-1 Radar and ICESat Laser Satellite Altimetry (Bamber et al., 2009), and MEaSUREs BedMachine 3.0 (Morlighem et al., 2020, Morlighem, 2022). These MEaSUREs inputs were chosen as they provide data at finer temporal resolutions, unlike the decadal averages generally provided by other velocity datasets available. Note that $knmax$ for

some years (2007, 2012, 2013) was unable to be calculated due to the lack of available surface velocity input data, which had insufficient spatial coverage. The *knmax* output data were then imported into QGIS and the passive ice boundaries for each year were visually traced for regions where *knmax* = 0.

To determine if changes in buttressing resulted in changes in the discharge across the grounding line, the ice discharge along the grounding line was calculated using the same velocity and thickness inputs as the *knmax* calculation. I assume plug flow (no bed-parallel vertical shear deformation), such that the surface velocities are the same magnitude as the basal velocities, and a normal vector to the depth-averaged velocities was computed for each segment of the grounding line (defined here as the line between two grounding line node points, where the node points are the interpolated straight lines that delineate purely floating from purely grounded regions of the ice shelf system). The normal vector is then multiplied by the velocity to calculate the dot product of a segment along the grounding line. The sum of these segments is multiplied by the ice thickness, density and length to calculate the flux across the grounding line: (Larour et al., 2012)

$$F_{Segment} = \rho_i H (V_x N_x + V_y N_y) L \quad (6)$$

$$F_{Total} = \frac{\sum F_{Segment}}{10^{22}} \text{ Gt/yr} \quad (7)$$

where V_i is the velocity in the x and y direction for that segment, N_i is the normal vector for that segment, and L is the length of that segment. The sum of the flux segments is divided by 10^{12} to convert the units into *Gt/yr*.

3.4 Decision-Making Framework

Recognising the critical role of in-situ observations in improving the understanding of the Cook Ice Shelf - WSB system, I apply the findings from the three research questions to construct a

framework that informs strategic deployment of observational instruments. Here, the decision-making framework is designed around the deployment of an instrument to measure in-situ ocean-driven melt - an Autonomous phase-sensitive Radio Echo Sounder (ApRES). ApRES is a ground-based radar instrument, and can autonomously measure vertical strain rates and basal melt rates through changes in internal layer depths, with millimetre depth precision (Nicholls et al., 2015, Young et al., 2018). This radar records a time series of amplitude and phase versus depth, enabling the tracking of internal layers through time (Cook et al., 2022). The decision-making framework uses information on melt, crevassing, and buttressing, as well as surface mass balance. These factors are normalised to a scale of 0 to 1, weighted, and multiplied together to produce an assessment of the spatial variation in suitability of deployment across the Cook Ice Shelf. Scores of 1 in this suitability term represent areas that are optimal for deploying ApRES; scores of 0 represent regions that are inappropriate to deploy ApRES. For basal melt rates, the lowest melt rate is scaled to a score of 0, and the highest melt rate is normalised to a score of 1, such that regions with higher values indicate optimal deployment sites, given that this framework targets high basal melt observations. The normalised crevasse factor is the calculated inverse of crevasse formation probability. Regions which were already crevassed on the surface ($\sim 100\%$ probability of crevasse formation) is given a score of 0, and regions which were least likely to crevasse ($\sim 0\%$ probability of crevasse formation) is assigned a score of 1. This considers the operational and data quality risks associated with deploying instruments in heavily crevassed regions, where structural instability and return radar signal scattering can compromise instrument safety and measurement accuracy. For normalising the buttressing, a piecewise function and interpolation is used. For regions seaward of the median passive ice boundary (i.e. passive ice) (calculated from 4.3), values of 0 is assigned. Using the median *knmax* geotiff, active ice regions ranging from 0 to 1 is directly assigned the same score ranging from 0 to 1, and regions with *knmax* values greater than 1 is assigned a score of 1.

However, active regions of the Cook Ice Shelf contained areas of tensile stress (due to internal ice shelf extension), which held negative $knmax$ values. These are still regions of active buttressing, so their score is interpolated (linear) to gap fill the area depending on the $knmax$ values of the adjacent regions. The deployment of ApRES instruments in regions of active buttressing is ideal, as it allows for the measurement of vertical strain thinning for regions that directly impact the acceleration and discharge of upstream grounded ice. Conversely, deployment in passive ice areas is avoided, because strain thinning in these regions do not necessarily translate to immediate impacts on inland flow. Regions of very high snowfall accumulation (defined as >0.6 m/year, assuming a 3-year deployment period for this case study) are non-optimal as the ApRES would likely be buried. Therefore, a threshold value of 0.6 m/yr surface mass balance was assigned, where values above this is scored 0, and values less than this are normalised between 0 and 1.

The four factors of basal melt rate, crevassing, buttressing, and surface mass balance were weighted and aggregated to calculate the combined decision-making framework score:

$$S_{Weighted} = (0.7S_M + 0.3S_B)(1.0S_B 0.8S_S) \quad (8)$$

where in Eqs (8), $S_{Weighted}$ is the final weighted score, S_M is the basal melt rate score, S_B is the buttressing score, S_C is the crevassing score, and S_S is the surface mass balance score. The first term $(0.7S_M + 0.3S_B)$ represents a weighted average of the basal melt rate and buttressing suitability scores, where the assigned weights (0.7 and 0.3) sum to 1, ensuring a balanced contribution between the two factors. As the aim of this framework is to prioritise ApRES deployment in regions undergoing intense ocean-driven melt, the basal melt rate factor was assigned a higher weighting (70%) relative to buttressing (30%). This weighting reflects the higher importance of capturing melt dynamics for this study objective, while still accounting for the structural influence of buttressing, but as a secondary consideration. A weighted average

score for these two factors is used, because a high melt score can compensate for a lower buttressing score, and a low score for either of these variables does not impact the safety of the instrument or quality of data (unlike crevassing and surface mass balance) but is instead weighted towards the aim of the scenario goals. Conversely, the second term ($1.0S_B0.8S_S$) places the crevassing and surface mass balance factors as multipliers instead of a weighted average. This means that if either of these factors drop towards 0, then the whole combined score also drops towards 0, even if melt and buttressing scores are high. Crevassing is given a multiplier value of 1.0, as it is an absolute limiting factor which directly negatively impacts the quality of ApRES data received and poses a risk of physical damage to the instrument. Surface mass balance is assigned a slightly lower multiplier value of 0.8 because its influence on ApRES deployment is more moderate: extremely low surface mass balance values (~ 0) already nullify the combined score regardless of the weighting, so the surface mass balance multiplier only affects intermediate accumulation rates (~ 0.5 to 0.6 m/yr). These conditions still impact instrument safety/logistics of deployment, but less severely than crevassing. All four factors are not calculated as a weighted sum (no multipliers) because this would allow one variable to inflate the overall score. For example, a site with low melt and weak buttressing could still rank highly if crevassing or surface mass balance conditions were favourable. This result would be misleading, as a given site would appear suitable despite the limited scientific value that would be collected (in relation to the chosen scenario of targeting regions undergoing intense ocean-driven melt).

The output of Eqs. (8) is then normalised to form a combined decision-making framework scaled from 0 to 1, where 1 represents ideal ApRES deployment locations. Note that the above decision-making framework for in-situ instrument deployment is one case study, chosen given the substantial uncertainties associated with basal melt of the Cook Ice Shelf (see section 5.1). The framework could be adapted for different instruments or target variables, as discussed in

more detail in section 5.4. Uncertainties and limitations associated with the above approach are also discussed in section 5.4.

Chapter 4 - Results

4.1 Melt Regime

To evaluate the first research question of quantifying the magnitude and mode of ocean-driven basal melting at the Cook Ice Shelf, I examine the spatial patterns in modelled basal melt rates and their relationship with local ocean conditions. The basal melt rate ranges from 0 to 15.4 m/yr, with highest values found near the grounding line (peaking at 15.4 m/yr, particularly throughout the main body of the Cook Ice Shelf), within 8 km inland of the 2006 ice front (7 - 11 m/yr), and throughout the central north-south body of the Cook Ice Shelf (7 - 10 m/yr) (figure 3a). The most prominent region of highest melt is located adjacent to the grounding line, near point *A* in figure 3a. This region is approximately 10 km in extent, with melt rates ranging from 12 to 15.4 m/yr. Conversely, regions of lowest basal melt rates are situated in the western body of the shelf, the inland regions near the grounding line of the far western end of the Cook Ice Shelf, and just east of the eastern shear margin (parallel between cross sections *B* and *C*). These low melt regions have melt rates of 1 - 2 m/yr. The mean basal melt rate of the Cook Ice Shelf is 4.0 m/yr. Figure 3b illustrates the mean and median basal melt rate anomalies of the entire ice shelf, calculated every three months from 2006 to 2018. The magnitudes of basal melt rate display approximately interannual variability, with the mean anomaly trendline reflecting a gradient of -0.04, and an r-squared value of 0.21. This suggests a weak decrease in basal melt rates over this period.

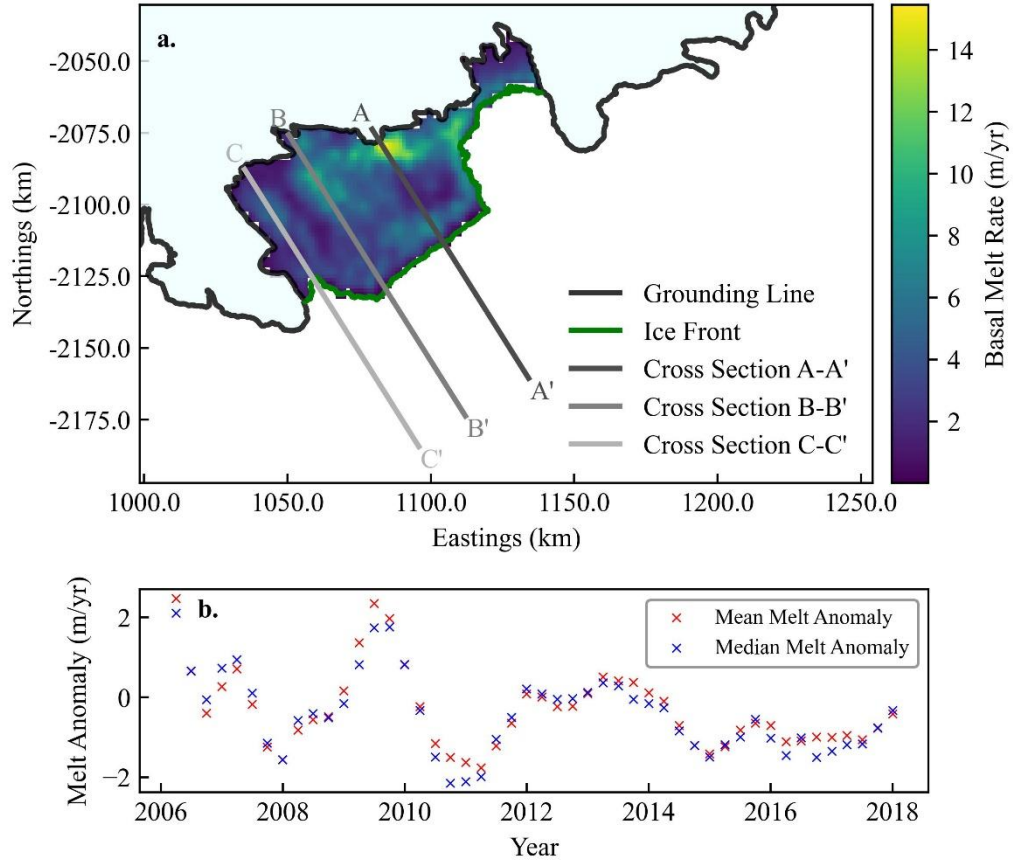


Figure 3: Basal melt rate of the Cook Ice Shelf (Richter et al., 2022), with three lines illustrating the cross-section segments. The 2003 grounding line is plotted as a solid black line (Bindschadler and Choi, 2011) and the 2011 ice front is plotted as a solid green line (Greene et al., 2024). (b): Mean and median basal melt rate anomalies plotted as a time series, from 2006 to 2018.

The three line of cross sections *A*, *B*, and *C* in figure 3a correspond to columns of *a*, *b*, and *c* in figure 4, from which I can examine and classify the water masses. All transects present similar ocean property patterns. Temperature (figure 4a-c) increases with depth, with coldest waters (-1.8°C) found near the ice front surface. There is generally an increase in temperature away from the ice shelf cavity. Practical salinity (figures 4d-f) increases with depth, with fresher waters ($\text{SU} < 34.30$) within 200 m from the ice shelf base, and saltier waters ($\text{PSU} > 34.50$) consistently at depths greater than 500 m below the surface. Neutral density (figure 4g-i) increases with depth/pressure, from 27.80 kg/m^{-3} closest to the upper ice front, to 28.14 kg/m^{-3} in the deepest regions around 100 km seaward of the Cook Ice Shelf grounding line. The neutral density gradient of 28.00 kg/m^{-3} is consistently situated at an approximate depth of 450 m. This is the isopycnal that is used to classify the difference between mCDW and ASW, with ASW defined

as having neutral density less than 28.00 kg/m^3 , and mCDW having neutral density values between 28 and 28.27 kg/m^3 . DSW, defined as neutral density values above 28.27 kg/m^3 , is not found anywhere in the domain considered here. A bathymetric sill present in the ice shelf cavity for each transect is a limiting factor for the incursion of mCDW. mCDW is present near the grounding line only in transect A; in transects B and C, where the bathymetry is shallower, limited to no mCDW is present. The uncertainties associated with the ice shelf cavity bathymetry are high, and this significantly impacts our confidence in the estimates of basal melt rates generated with WAOM for the bathymetry used, as discussed in more detail in section 5.1.

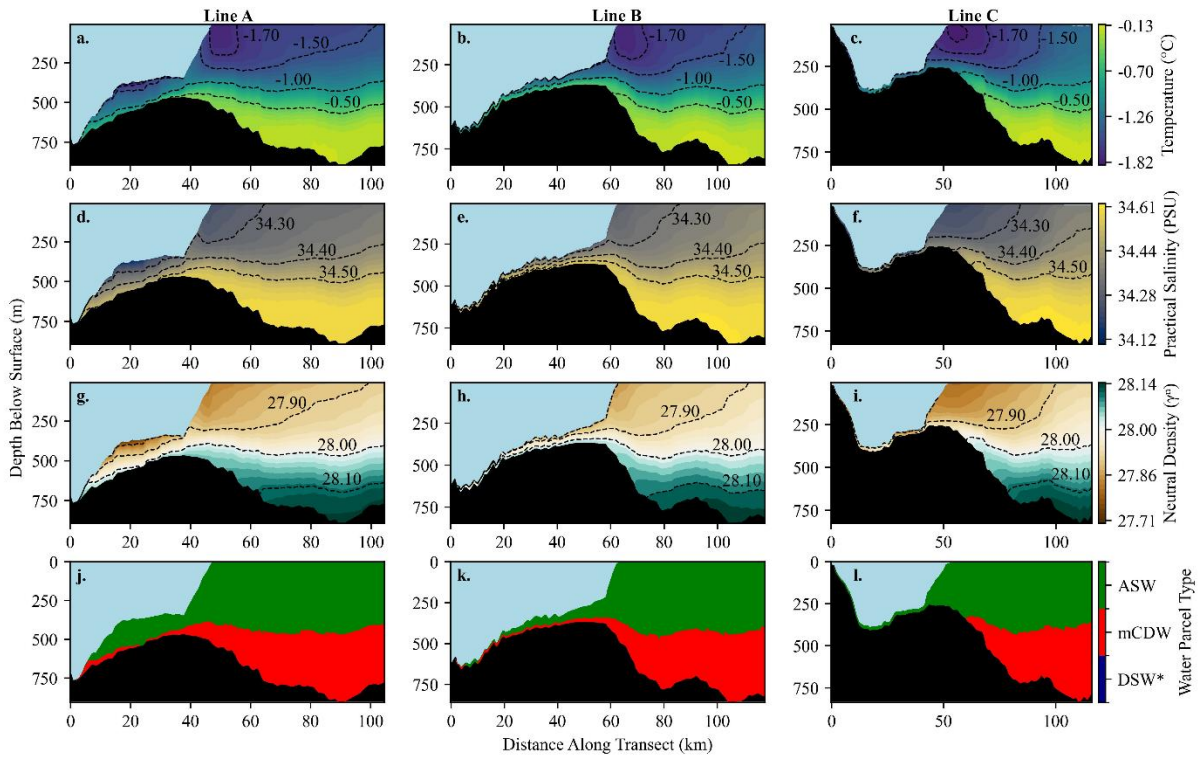


Figure 4: Cross section segments (Lines A-C) aligned to their respective column, displaying potential temperature ($^{\circ}\text{C}$), practical salinity (PSU), neutral density (kg/m^3) and water mass types (Richter et al., 2022). Note that the X and Y axis scales differ slightly between each cross section.

There is a clear positive relationship between basal melt rates, bathymetry, ice draft depth, and the proximity to the grounding line (figure 5). High basal melt rates correlate with deeper bathymetry. These regions are generally located closer to the grounding line, although isolated regions further from the grounding line (in the vicinity of the ice front) also show elevated melt rates.

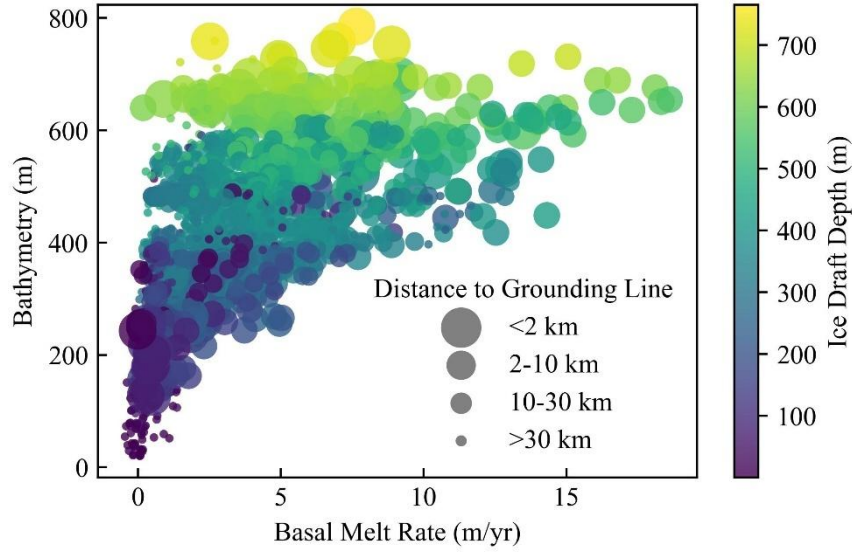


Figure 5: Scatter plot of the monthly mean basal melt rates and bathymetry of the Cook Ice Shelf at 2 km horizontal resolution, from July 2011 WAOM outputs (Richter et al., 2022). Data point colour is according to ice shelf draft depth. Size of data points is according to the proximity to the grounding line (Bindshadler and Choi, 2011). Note that bathymetry and ice shelf draft depth is in metres below sea level.

4.2 Crevassing

To respond to the second research question of identifying which regions of the Cook Ice Shelf are most vulnerable to crevasse formation, I first consider the patterns in strain rates across the shelf (figure 6a). The $\dot{\epsilon}_{eff}$ is generally greatest along the grounding line (0.02 to 0.07 a^{-1}), in the far western region of the Cook Ice Shelf, and along the eastern shear margin (0.02 to 0.1 a^{-1}). The rest of the Cook Ice Shelf has an $\dot{\epsilon}_{eff}$ an order of magnitude smaller (i.e. generally less than 0.01 a^{-1}). Crevassed regions are found in areas in high strain rates (generally above 0.02 a^{-1}), but also extend westwards of the eastern shear margin, and northwards of the crevassed field found in the western body of the Cook Ice Shelf.

The PDF curves in figure 6b map the probability density for each $\dot{\epsilon}_{eff}$ value (i.e. how likely it is to find a region of ice with a certain $\dot{\epsilon}_{eff}$ value on the Cook Ice Shelf for a given year). Note that figure 6b shows probability densities not probabilities, and as such, a value of 100 does not mean 100% probability of all regions having a $\dot{\epsilon}_{eff}$ value of 0.006 a^{-1} , but instead that this $\dot{\epsilon}_{eff}$ value is *most common*. Both the crevassed regions and those regions free from crevasses show $\dot{\epsilon}_{eff}$ values of 0.004 to 0.008 a^{-1} - these $\dot{\epsilon}_{eff}$ values are the most prevalent ranges across the Cook

Ice Shelf. A threshold value of 0.018 a^{-1} marks where the PDFs for crevassed regions begin to exceed the annual PDFs for the entire Cook Ice Shelf. This means that areas with an $\dot{\epsilon}_{eff}$ values above 0.018 a^{-1} are more likely to be crevassed, although some crevasses still occur in regions with lower strain rates. The skewness of the PDFs suggests that $\dot{\epsilon}_{eff}$ are increasing and spreading across the ice shelf, indicating a general rise in extensional and shear stresses throughout the Cook Ice Shelf.

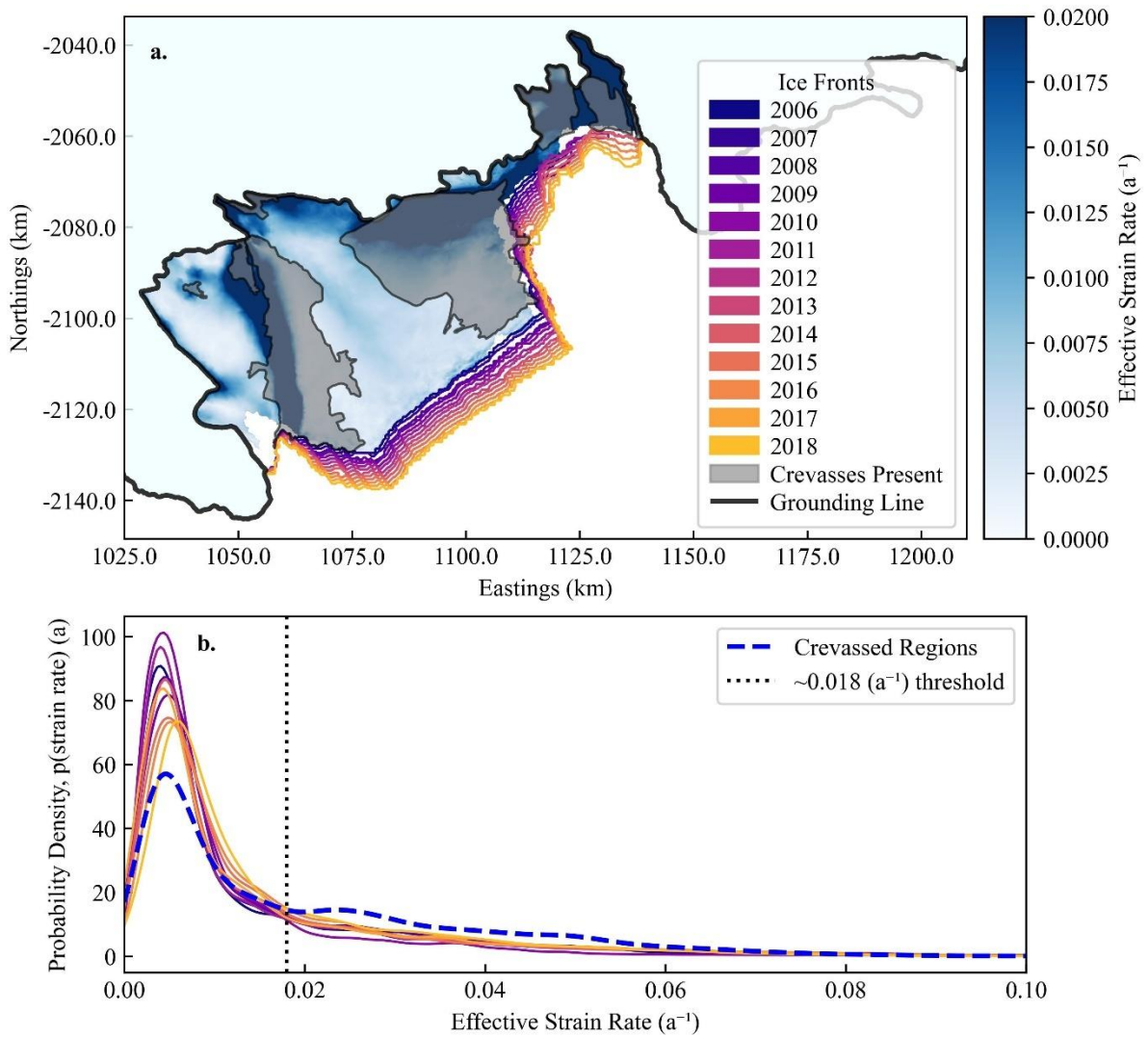


Figure 6: (a) Median effective strain rates of the Cook Ice Shelf from 2006 to 2018, overlaid with grey areas illustrating regions which are already crevassed, from 2008 satellite imagery (Bindshadler et al., 2008). The 2003 grounding line is plotted as a solid black line (Bindshadler and Choi, 2011) and annual ice fronts from 2006 to 2018 are overlaid in colours (Greene et al., 2024). (b) Probability density functions (PDF) encompassing effective strain rate for the entire ice shelf for each year are plotted as solid lines, corresponding to the same colours in (a). The PDFs for the crevassed area of median strain rate is overlaid as a dotted blue line. A vertical dotted line is overlaid, representing the threshold strain rate where the crevassed PDF crosses over the PDFs of the entire ice shelf.

Figure 7a shows the conditional probability of crevasses forming, given the $\dot{\epsilon}_{eff}$ value for that region. This is calculated from the median $\dot{\epsilon}_{eff}$ values for the areas which are already crevassed (Eqs. 2). Crevasses are most likely to form throughout the eastern shear margin, within 5 km of the grounding line (60 to 80% likelihood), and in the far west (>80%). The body of the Cook Ice Shelf is least vulnerable to crevasse formation, with probabilities falling between 20 to 35%. In general, surface crevasses detected fall within regions defined as having a high likelihood of crevassing (from the strain rate calculation), although there are some regions where crevasses are present, despite the lower likelihood (e.g. probability ~30%).

The Cook Ice Shelf ice shelf front consistently increased in extent from 2006 to 2008 (figure 7b and table 3 [appendix A]) at an average rate of 36.50 km² per year. Note that this was calculated assuming a constant grounding line, but a changing ice front.

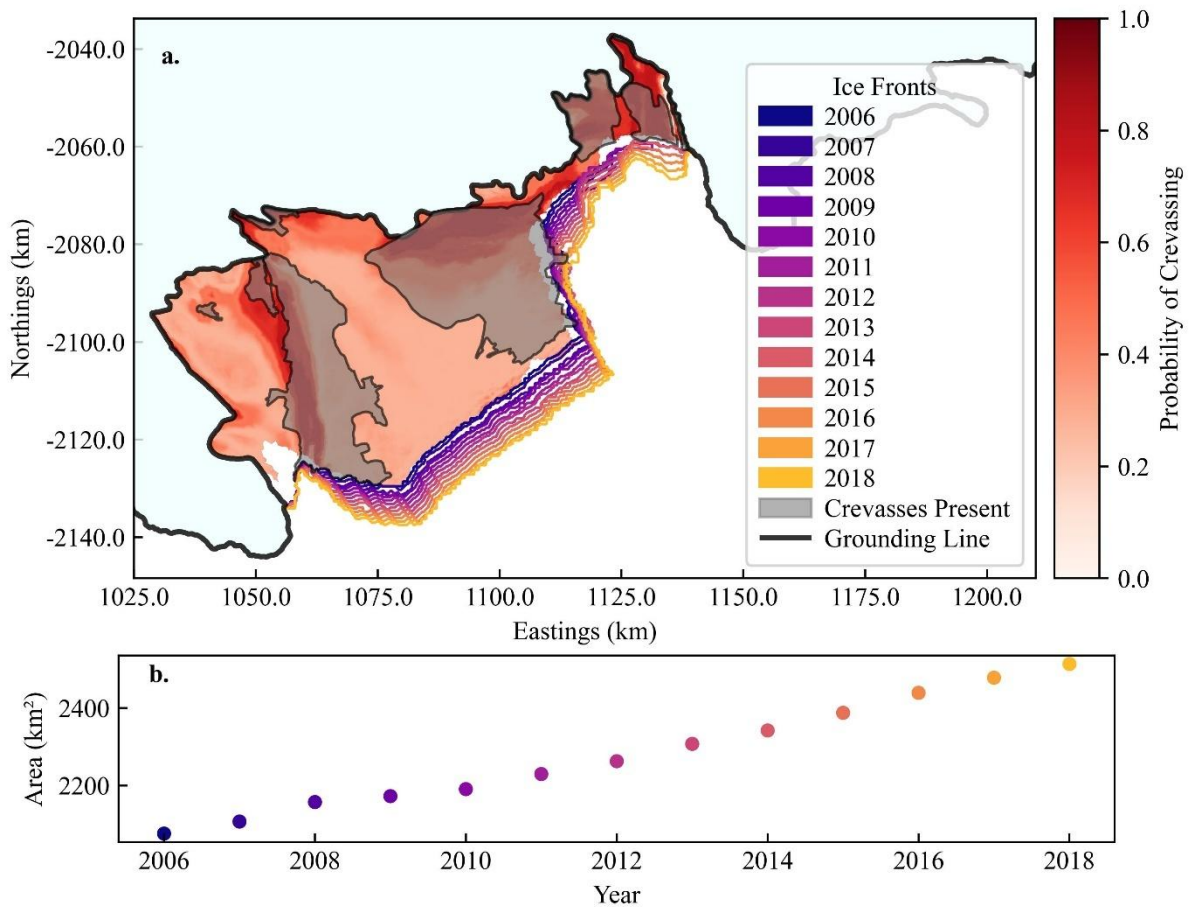


Figure 7: (a) Visualisation of the probability of crevasses occurring throughout the Cook Ice Shelf from 2006 to 2018, overlaid with grey areas illustrating regions which are already crevassed, based off 1999-2003 satellite imagery (Bindshadler et al., 2008). The 2003 grounding line is plotted as a solid black line (Bindshadler and Choi, 2011) and annual ice fronts from 2006 to 2018 are overlaid in a plasma colourway (Greene et al., 2024). **(b)** The Cook Ice Shelf absolute area is plotted as a time series.

4.3 Buttressing

To address the third research question of assessing changes in the buttressing capacity of the Cook Ice Shelf, I first consider the evolution of the passive ice boundary (figure 8a). The boundaries bisect the Cook Ice Shelf along an approximate east-west axis, with a sharp angle towards the central western ice front. Passive ice boundaries remain relatively consistent between 2006 and 2018, except in 2018 when the boundary extends seaward along the eastern and western margins of the Cook Ice Shelf relative to other years. Passive ice boundaries coincide with velocities of 680 to 720 m/yr, except at their southernmost circular portion near the grounding line, where velocities decrease to 570 m/yr. The time series of grounding line flux from 2006-2018 is shown in figure 8b. The flux during this period is relatively constant, decreasing from 0.35 Gt in 2006 to -1.69 Gt in 2015, then slightly increasing to -0.73 Gt in 2018 (table 3 [appendix A]). The transition from positive flux in 2008 to negative flux in 2009 indicates the shift from a net ice loss to a net ice gain. Note that velocity data in 2007, 2012 and 2013 was of too low quality to enable calculation of the discharge during these years, although I do not expect any substantial deviations from the trends analysed through the rest of the period.

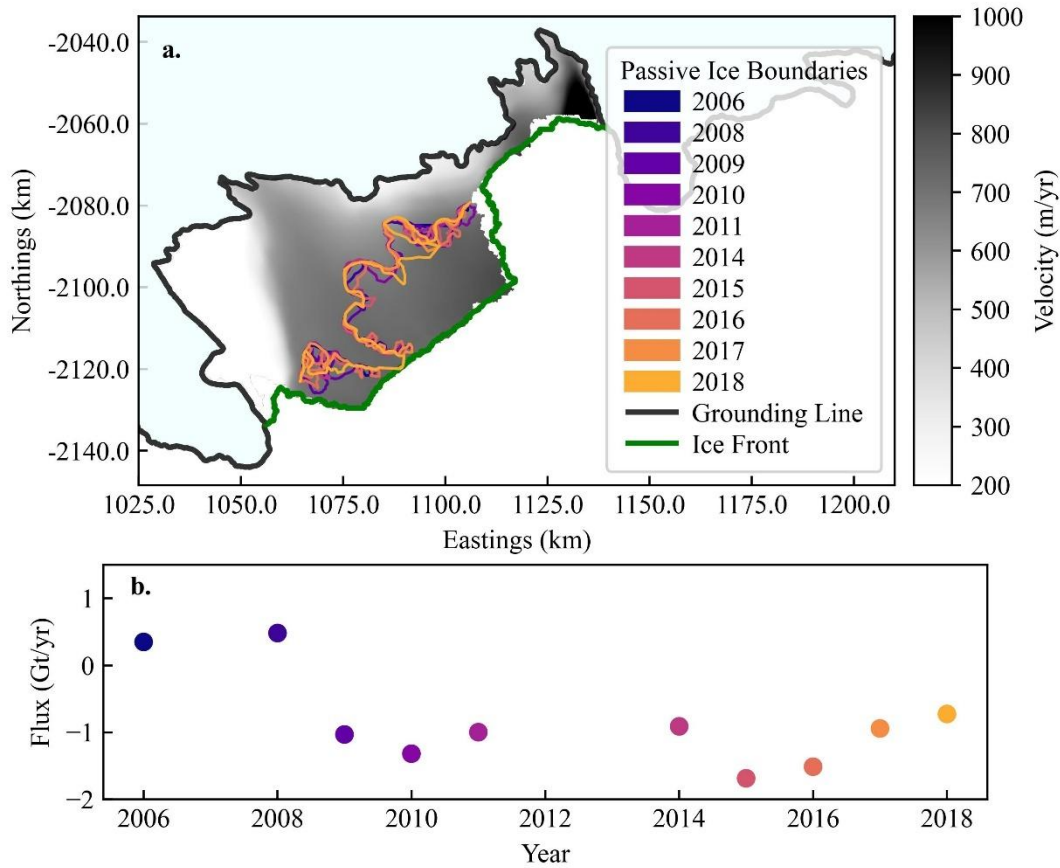


Figure 8: (a) Visualisation of the mean surface velocities of the Cook Ice Shelf from 2006 to 2018 (Mouginot et al., 2017), with the 2003 grounding line overlaid as a solid black line (Bindshadler and Choi, 2011) and the 2006 ice front overlaid as a solid green line (Greene et al., 2024). The annual passive ice boundaries are plotted in purple and orange lines. (b) The Cook Ice Shelf flux is plotted as a time series. Note that the years 2007, 2012 and 2013 are excluded due to insufficient input data.

Chapter 5 - Discussion

This study aims to assess the vulnerability of the Cook Ice Shelf to climate change through changes in buttressing in response to ocean-driven melt and crevassing. The results find that basal melting is dominated by mCDW, with melt - although low - is amplified by bathymetric access to sub-ice-shelf cavities. Crevasses are concentrated near the grounding line and the eastern shear margin, but currently act as local hazards rather than system-wide destabilisers to buttressing. Similarly, minimal calving over the period analysed here indicates that calving has had little impact on Cook Ice Shelf stability in recent decades. Buttressing remains stable across the southern half of the shelf, despite local highs in basal melt rates and crevassing, suggesting

little evidence of a current response to climate change. The following sections discuss these results in detail, considering their consistency with other Antarctic settings, associated uncertainties, and broader implications for future vulnerability to climate change. From this, I then highlight where our results are most uncertain, using this to define our approach to targeting in-situ measurements via the ApRES decision-making framework.

5.1 Melt Regime

The magnitude of basal melt across the Cook Ice Shelf is presently low, with melt concentrated near the grounding line but little evidence of widespread thinning, suggesting that the shelf is currently at low vulnerability to ocean-driven thinning. The average basal melt rate of ~ 4.0 m/yr for July 2011 from the WAOM (Richter et al., 2022) is higher than literature, such as basal melt rate outputs from Cryo-Sat 2 (Davison et al., 2023, Adusumilli et al., 2020) which state an average of 2.69 m/yr for this same month, but with an error of up to ± 3.45 m/yr. However, direct comparison between these datasets is limited by notable uncertainties in both approaches. For WAOM, uncertainties arise from errors in the sub-ice-shelf cavity geometry in Bedmap2, which influences basal melt distribution. High-resolution satellite estimates of ice shelf-ocean interaction remain poorly constrained (as reflected by the high error values up to 3.45 m/yr), particularly near grounding lines, making it difficult to confidently compare melt rates. The basal melt rate of the Cook Ice Shelf is relatively low compared to West Antarctic and Antarctic Peninsula ice shelves. For example, the Thwaites ice shelf in the Amundsen sector consistently melts at rates between 10 to 30 m/yr (Davison et al., 2023). Furthermore, the basal melt rate anomaly time series suggests that from 2006 to 2018, there is no clear trend of potential consistent increases in melting. These results suggest that ocean water masses beneath the Cook Ice Shelf have not yet been strongly influenced by climate change, unlike those in West Antarctica. While localised regions of high basal melt indicate areas of potential vulnerability,

the generally low and spatially limited melt rates imply that the Cook Ice Shelf is not currently undergoing significant dynamic ocean-driven thinning.

The dominant modes of melt at the Cook Ice Shelf are associated with mCDW and ASW. Higher melt rates and more rapid thinning is generally associated with mCDW (Adusumilli et al., 2020) as it has a relatively high water temperature. This is the dominating mode in the Cook Ice Shelf. Conversely, ASW is cooler, fresher, and less dense than mCDW (Ribeiro et al., 2023). This water parcel type typically resides closer to the surface, and primarily contributes to melt near the ice front (McCormack et al., 2024). These findings are only partly consistent with the limited literature in this understudied region, where the presence of mCDW has been suggested (Constantino and Tinto, 2023), but ASW has not been reported. The closest available oceanographic data dates back to 1957-58, where there was no evidence of mCDW, and no recent ship-based conductivity-temperature-depth surveys exist (Constantino and Tinto, 2023). This highlights the considerable uncertainties in ocean properties near the Cook Ice Shelf, and underscores the importance and need to constrain the role of ocean water masses in driving basal melt in this region.

While the two melt modes of mCDW and ASW help explain the spatial variability in basal melt rate, the magnitude and distribution of basal melt are primarily controlled by cavity geometry and poorly constrained bathymetry, as is expected given the reliance of ocean model output in this study. In many Antarctic settings, basal melt rates are higher where bathymetry allows warmer water parcels such as mCDW to reach the grounding line (Nicola et al., 2025). However, at the Cook Ice Shelf, a bathymetric sill appears to restrict the supply of mCDW, particularly along the transects considered here. While there is some mCDW-driven melt along transect *A* (figure 4) the bathymetry here, and throughout the region is highly uncertain. That is, the available bathymetric data is associated with uncertainties of up to ~320 m (Fretwell et al., 2013). This uncertainty arises because sub-ice-shelf cavity geometry is typically

interpolated between the grounding line and the ice front. This is a common approach in Antarctic datasets (Morlighem et al., 2020, Morlighem, 2022), but could lead to large uncertainties. Oceanographic data of temperature and salinity is also poorly resolved, as these data inputs originate from model outputs, with no physical ship in-situ measurements available. This limits the confidence in attributing melt drivers and the magnitude of basal melt rates. Addressing this gap is particularly important because basal melting is both the primary driver of Antarctic ice loss (Davison et al., 2023) and one of the least constrained processes, reinforcing the need for targeted observations. Accordingly, this knowledge gap motivates the deployment of a decision-making framework to deploy ApRES instruments on the Cook Ice Shelf, as discussed in section 5.4.

5.2 Crevassing

The distribution of crevasses across the Cook Ice Shelf reflects a strong relationship with effective strain rates, with distinct thresholds indicating where crevassing is most likely to occur. Crevasses are generally found where $\dot{\epsilon}_{eff}$ is higher. The damage from these crevasses makes these regions more vulnerable to structural weakening (reduced buttressing), faster ice flow, and hence greater ice loss (Li et al., 2024). Crevasses typically form when the strain rates exceed the fracture toughness of the ice (Thompson et al., 2020). This corresponds with literature, where crevasse initiation is linked to strain rates ranging from 0.001 to 0.163 a^{-1} (Campbell et al., 2013). The PDFs of $\dot{\epsilon}_{eff}$ further highlight this relationship, illustrating that most of the Cook Ice Shelf clusters around values of 0.004 to 0.008 a^{-1} , but that a threshold of $\sim 0.018 \text{ a}^{-1}$ marks the point at which crevassed regions become more frequent than uncrevassed areas, suggesting a link between $\dot{\epsilon}_{eff}$ and fracture formation. These results emphasise that $\dot{\epsilon}_{eff}$ is the dominant control on present-day crevassing patterns across the Cook Ice Shelf, with clear thresholds increasing the shelf vulnerability to crevassing.

Although $\dot{\epsilon}_{eff}$ across the Cook Ice Shelf are increasing and spreading spatially, the absence of major calving events suggests that crevassing remains a localised hazard, rather than a system-wide destabiliser. This interpretation is supported by the steady expansion of the Cook Ice Shelf from 2006, during which ice extent increased by an average of 36.5 km² per year, with the absence of any major calving events. The PDFs of $\dot{\epsilon}_{eff}$ (figure 6b) reveal a slight increase in skewness and standard deviation, indicating that strain rates are not only increasing, but also spreading more broadly across the Cook Ice Shelf, reflecting a growth in extensional and shear stresses. This generally suggests structural weakening and accelerating flow (Ranganathan et al., 2025) which may manifest into future calving events, highlighting the need to map crevasses in greater detail and to be monitored carefully to predict potential changes in shelf buttressing and future dynamic responses. Although surface crevasses can pre-condition ice shelves and act as predictors for hydrofracture and calving (Surawy-Stepney et al., 2023), the current stress increases are yet to translate into large-scale strain failure at the Cook Ice Shelf, indicating that the vulnerability state of this shelf to climate change is currently sufficiently stable.

Although the crevassing analysis provides insight into the vulnerability of the Cook Ice Shelf to climate change, several data and methodological limitations constrain the certainty of these interpretations. The mapping of crevassed regions is based on the 2008 Landsat satellite imagery (Bindshadler et al., 2008), where only visible surface crevasses were traced - this excludes the presence of subsurface and basal crevasses. Assuming there are hidden crevassed fields not identified in the satellite imagery, the PDF curves may shift, and the conditional probability of crevasse formation would be biased. This could underestimate the vulnerability of the Cook Ice Shelf to crevassing susceptibility to future calving. Furthermore, the analysis does not take into account processes such as meltwater propagation - when surface crevasses or fracture structures are filled with water, and cause hydrofracture that can lead to mechanical weakening (Benn et al., 2007), further rifting, and the expansion of crevasses. Despite this, the

calculated $\dot{\epsilon}_{eff}$ map inputs are sourced from ITS_LIVE surface velocity datasets (Mouginot et al., 2017), and are robust. There is also some uncertainty in the ice shelf area extent, and its variability over time. The annual extent from 2006 to 2018 utilises the same grounding line (Bindshadler and Choi, 2011) for each year. This was necessary because annual grounding line datasets are generally unavailable; most published datasets only provide multi-year intervals (2003, 2009, 2018). This low frequency is due to limited satellite coverage, reinforced by the need for radar or altimetry to accurately detect the transition from a grounded ice sheet to floating ice shelf. Although the use of a singular grounding line input introduces some uncertainty in where grounding line migration may have occurred, it also provides a consistent spatial reference that allows for accurate comparison of annual ice front positions and crevasse evolution. This supports the reliability of observed trends in ice extent, even if the absolute magnitudes of change may be slightly underestimated. Despite these constraints, the spatial patterns identified - especially along the eastern shear margin, near the grounding line, and in the west - highlight critical zones of structural weakness. These results suggest that while the Cook Ice Shelf is locally weakened, the overall stress regime remains relatively stable under current climate forcings.

5.3 Buttrressing

The spatial distribution of active and passive ice across the Cook Ice Shelf portrays a stable buttressing regime, with limited signs of system-wide dynamic change between 2006 and 2018. The mapped passive ice boundaries bisect the lower half of the Cook Ice Shelf, indicating that the lower northern portion of this shelf is passive and lacks contribution to resistive buttressing forces, whereas the upper southern half of the shelf is dynamically active and provides stabilising forces against the upstream Cook Glacier. Ice velocities reflect this division/pattern, with faster flow found in the northern and western regions, where extensional tensile stresses are highest. The annual flow and thus kn_{max} calculations are dependent on surface velocity

datasets (Mouginot et al., 2017), and are robust (similar to the $\dot{\epsilon}_{eff}$ calculations). However derived metrics such as kn_{max} and passive ice boundaries depend on gap fill interpolation of surface velocities, which may smooth over localised stress concentrations and underestimate vulnerability in some regions. The minor variations in stress and flow patterns correspond to minor fluctuations in the position of the passive ice boundary over time. The spatial patterns of basal melting also partially align with this flow-buttressing structure. Higher basal melt rates near the ice front link to the northern passive ice regions, but these regions of high melt near the grounding line do not correlate to signs of weakened buttressing in this area. This indicates that while basal melt rates influence the Cook Ice Shelf surface mass balance, it is yet to translate into substantial reductions of buttressing or landward advancement of the passive ice boundary. However, current melt and ice shelf thinning could push the system past a critical threshold, where the ice shelf is no longer able to effectively restrain upstream grounded ice. Crossing this threshold could result in the increased flow of upstream grounded ice speed and hence the contribution of ice sheets to global sea level rise (Davison et al., 2023, Gudmundsson et al., 2019). Overall, these findings suggest that the Cook Ice Shelf currently maintains a balance between flow, stress, and melt-driven thinning, with only minor localised variations in the annual passive ice boundaries and buttressing strengths.

The absence of major calving events and a sustained negative outflux suggests that although crevassing is locally extensive, it has not yet translated into significant loss of buttressing across the Cook Ice Shelf. The negative outflux from 2009 onwards represents a net ice gain, likely reflecting increased surface mass balance from snowfall accumulation and suggesting overall thickening or lateral expansion. This suggests the net thickening of the Cook Ice Shelf from 2009 to 2018, despite ongoing basal melt. Although surface crevasses are widespread along the eastern shear margin, throughout the west and near the grounding line, the spatial context of active ice in these same regions, coupled with a net ice gain, suggests that these crevasses have

not yet substantially reduced the system-wide buttressing capacity of the active ice shelf. This suggests that crevasses pose more of a localised hazard for stability and instrument deployment, rather than a large-scale loss of buttressing. Furthermore, the absence of major calving events throughout this period reinforces this interpretation, suggesting that the Cook Ice Shelf has retained sufficient buttressing capacity to offset the weakening effects of basal melt and crevassing. This absence of a dynamic response hints that the vulnerability of the Cook Ice Shelf to climate change is yet to escalate.

5.4 Decision-Making Framework

The decision-making framework developed for this thesis case study centres basal melt rates as the primary target variable. Given the detection of mCDW in this system (section 4.1), and projections that background ocean warming under climate change will increase mCDW supply in the future (Jordan et al., 2023), basal melt is potentially the driver of greatest future change. The results (sections 4.1 - 4.3) highlight that the Cook Ice Shelf has experienced minimal changes in calving extent, strain rate, crevassing, and grounding line outflow discharge. However, melt rates are highly uncertain due to the sparsity of observations from satellite remote sensing and model estimates. This uncertainty is critical, as the melt regime shifts observed in other Antarctic systems have showcased that basal melt rates can rapidly change over relatively short periods of years (Pritchard et al., 2012, Rintoul et al., 2016). It is essential to have a robust way to monitor changes in basal melt rates (such as the magnitude and spatial variability), so the melt regime process can be identified to improve the prediction and understanding of future changes. Therefore, basal melt rates are targeted as the central parameter of this framework, as it represents the greatest source of uncertainty and offers the greatest potential to improve the understanding of ice-ocean interactions at the Cook Ice Shelf, thereby increasing confidence in predicting the future response of this shelf to climate change.

Figure 9 displays the four normalised factors of basal melt rates, crevasses, buttressing, and surface mass balance. Regions of high basal melt rates (high suitability score) are located near the grounding line and ice front, and through the north-south body of the Cook Ice Shelf. Lower crevasse factor scores are naturally found in crevassed regions, and high factor scores are found throughout the north-south body of the shelf, and in the far east. High buttressing factor scores are found throughout the upper active/dynamic ice region, and low scores in the lower half of the Cook Ice Shelf, and throughout the far west (absent of *knmax* output). For surface mass balance, high factor scores are prevalent throughout the north-south central body of the Cook Ice Shelf, and low factor scores are applicable in the eastern and far western regions.

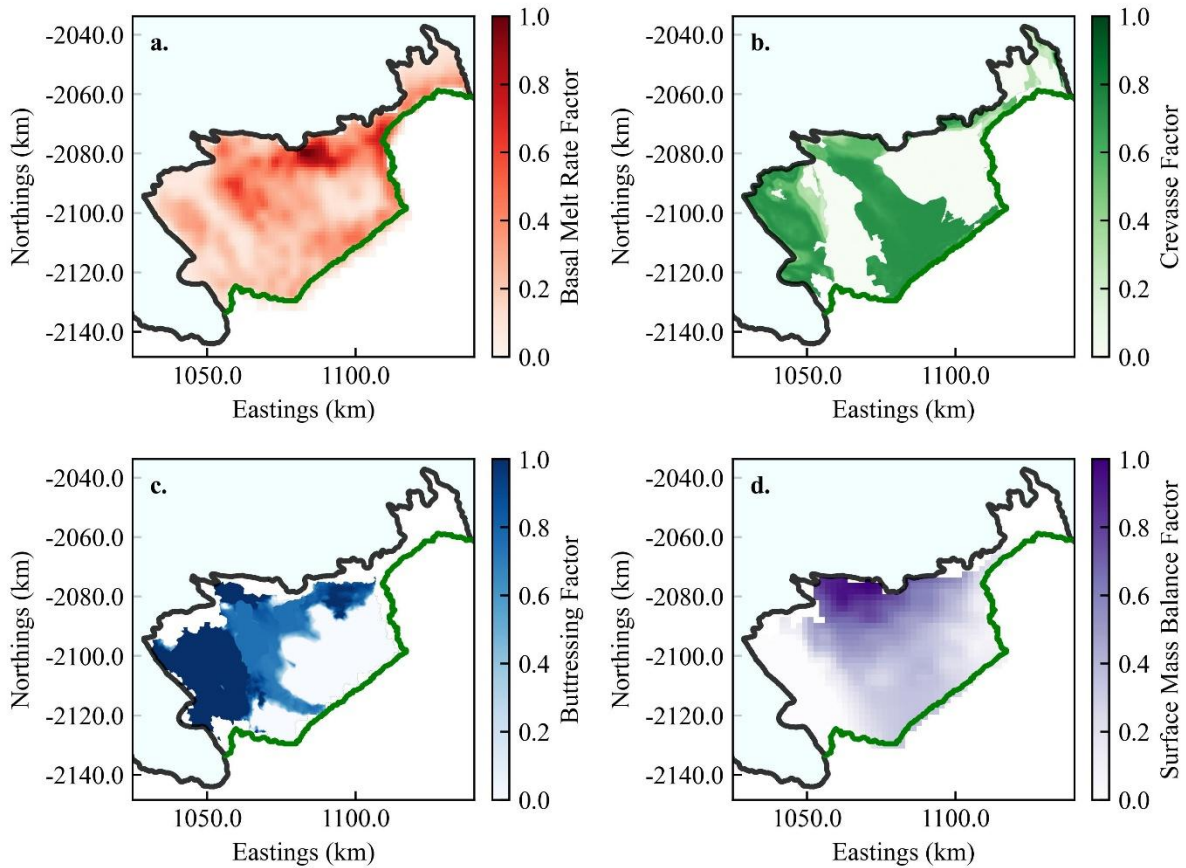


Figure 9: Visualisation of the normalised factor influence attributed to (a) basal melt rates (Richter et al., 2022), (b) crevasses (Bindshadler et al., 2008), (c) buttressing, and (d) surface mass balance (Paolo et al., 2024) overlaid with the 2003 grounding line plotted as a solid black line (Bindshadler and Choi, 2011) and the 2006 ice front plotted as a solid green line (Greene et al., 2024).

Figure 10 combines the four factors represented in figure 9 into a single combined, weighted suitability score for the deployment of ApRES which targets the surveyance of regions

undergoing intense ocean-driven melt. Ideal regions for instrument deployment are characterised by low basal melt rates, low probability of crevassing, active ice buttressing, and low surface mass balance. These suitable regions are identified in the southern-central region of the Cook Ice Shelf, stretching ~ 20 km inland of the grounding line. Conversely, the bulk of the Cook Ice Shelf remains unsuitable for ApRES deployment. This is influenced by crevassing in the western body and eastern shear margin, high surface mass balance in the far west and east, and passive ice in the northern lower half of the Cook Ice Shelf.

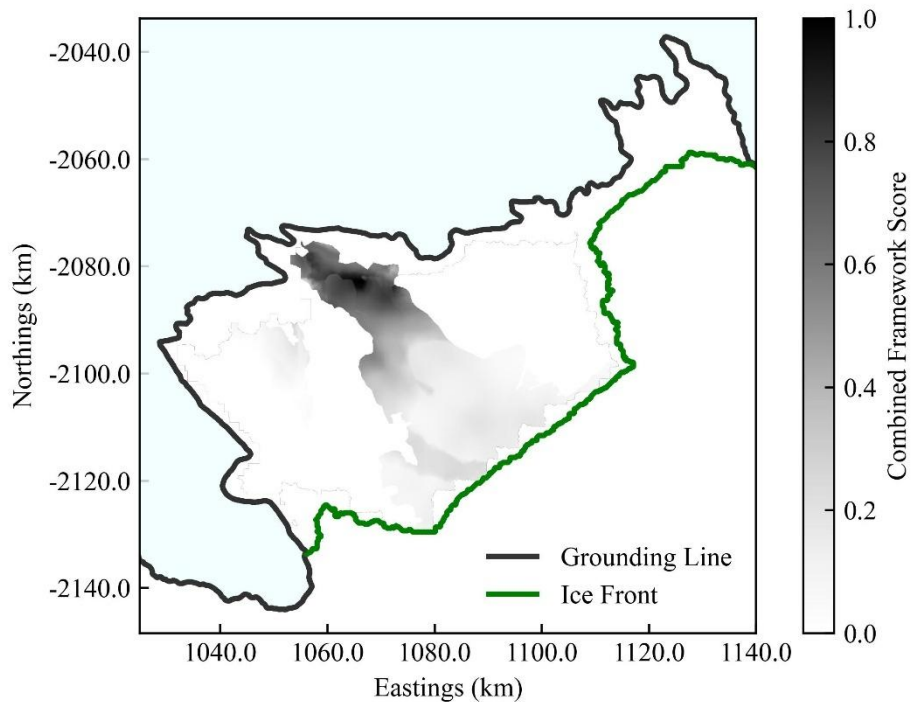


Figure 10: The combined framework addressed for targeting regions of the Cook Ice Shelf undergoing intense ocean-driven melt, overlaid with the 2003 grounding line plotted as a solid black line (Bindshadler and Choi, 2011) and the 2006 ice front plotted as a solid green line (Greene et al., 2024).

By targeting regions of high basal melt rates identified near the grounding line and throughout the central north-south body of the Cook Ice Shelf (section 4.1), vertical strain thinning and melt rate measurements can be used to better understand ice-ocean interactions, improve melt parameterisations, and reduce uncertainties in projections of Antarctica's future mass loss. These areas exhibit the strongest influence of mCDW and hence represent the zones of greatest melt-induced thinning. Within this framework, crevassing throughout the western sector and

the eastern shear margin (section 4.2) acts as a hard constraint, regardless of dynamical importance (in relation to active or passive ice), as fractured ice features can scatter and distort radar signals, reducing data quality, and pose operational risks to instruments. Active buttressing regions in the southern half of the Cook Ice Shelf (section 4.3), where the ice shelf exerts resistive stresses on the upstream ice sheet, are ideal targets for ApRES because they allow for direct measurement of vertical strain rates and thinning processes that directly influence upstream ice. These areas are dynamically important, as any weakening in their structure could propagate inland towards the grounding line, increasing discharge. Conversely, passive ice found in the southern half of the Cook Ice Shelf carries lower deployment priority, as these areas provide little to no buttressing, and could theoretically be removed without causing increases in upstream ice flow discharge (Fürst et al., 2016). Similarly, regions of low surface mass balance throughout the central body of the Cook Ice Shelf are preferred, as high snowfall rates can bury ApRES instruments, rendering them buried and irrecoverable.

There are a number of limitations surrounding my approach to the development of the decision-making framework for ApRES deployment on the Cook Ice Shelf. First, basal melt rate estimates have moderate spatial and temporal resolution (2 km, 3 months) but are sparsely validated by a lack of in-situ measurements. This limits the confidence in quantifying the accuracy of melt rate estimates, and introduces uncertainties into the smaller scale analysis and interpretations of spatial variability in basal melting, due to these coarse resolutions. Second, crevasse mapping relies on 2008 Landsat satellite imagery (15 m), which captures surface but not subsurface gestures. This limits the ability to assess temporal variability in crevassing and may not reflect current ice shelf conditions. Consequently, this may falsely represent the present-day physical risks for field deployment safety, and the influence of ice shelf mechanical weakening on the vulnerability to crevasse formation. In contrast, some datasets are more robust: $\dot{\epsilon}_{eff}$ and kn_{max} values are derived from surface velocity inputs, and surface mass balance

data are sourced from MEaSURES ITS_LIVE estimates (but at 1.9 km spatial resolution) (Paolo et al., 2024). However, these spatially coarser datasets can mask fine-scale spatial variability, potentially smoothing out critical localised features such as high strain concentration regions. Uncertainties in the derivation of velocity can propagate directly into the calculation of strain and buttressing, which can result in the bias of the identification of dynamically active versus passive ice. Furthermore, the coarse resolution of surface mass balance data may mask small, high accumulation regions, potentially leading to the misplacement of instruments in localised zones of unusually high burial rates. These limitations collectively result in uncertainties in the spatial weighting of each factor of this decision-making framework (basal melt rate, crevassing, buttressing, surface mass balance). As higher-resolution datasets become available through coordinated international efforts, these framework uncertainties can be reduced. Additionally, other variables such as wind exposure and meltwater lakes are omitted, as their contribution to overall melt is expected to be minimal. QuikSCAT satellite observations indicate that the Cook Ice Shelf experienced a mean surface meltwater flux of approximately 78.47 mm weq/yr (water equivalence) between 1999 and 2009 (Trusel et al., 2013), which is negligible compared to the mean basal melt rate of 4.0 m/yr calculated for July 2011 (Richter et al., 2022).

This process-based framework can also be adapted for other Antarctic ice shelves or objectives by adjusting factor weightings, thereby offering a transferable and scalable approach to instrument deployment. This reduces uncertainties and strengthens future ice modelling efforts centred around shelf vulnerability to climate change. For example, similar approaches could be used to inform the deployment of in-situ instruments associated with *Antarctica InSync* - a multinational programme launched under UNESCO's "UN Decade of Ocean Science for Sustainable Development (2021-2030)" campaign (InSync, 2025). This program aims to synchronise circumpolar observations to improve understanding of Antarctica's role in the global system as climate change progresses, and also plans to survey the Cook Ice Shelf directly

as part of this program. ApRES-based monitoring of basal melt rates and strain thinning across dynamically significant regions like the Cook Ice Shelf aligns directly with InSync's mission to generate high-resolution datasets that inform and support global climate projections and sea-level rise assessments. In summary, the decision-making framework developed above shows promise in bridging local-scale process studies with international efforts to synchronise Antarctic observations, supporting the understanding of the Cook Ice Shelf vulnerability to future climate change.

Chapter 6 - Conclusion

The overarching aim of this thesis was to assess the present-day vulnerability of the Cook Ice Shelf to climate change, through reduced buttressing in response to ocean-driven melt and crevassing. This was addressed through three research questions: (1) what is the magnitude and mode of ocean-driven melt at the Cook Ice Shelf, (2) what regions are vulnerable to crevasse formation, and (3) how has the Cook Ice Shelf flow regime evolved and what does this reveal about changes in the buttressing capacity. Basal melt was found to be dominated by mCDW near the grounding line and ASW near the ice shelf front. However, the magnitude of melt and cavity geometry are highly uncertain, highlighting the need for concentrated observational campaigns in this region. Surface crevasses were already present throughout the west and eastern shear margin, and were most likely to form near the grounding line. Despite this, ice extent steadily increased over time, and there is no evidence of the impact of crevassing on increased susceptibility to calving. Active buttressing is found throughout the southern half of the Cook Ice Shelf, and passive ice with no buttressing trends throughout the northern half. Buttressing did not change dramatically between 2006 and 2018, indicating that the Cook Ice Shelf is not currently undergoing large dynamic changes. Collectively, these results suggest that

although localised vulnerabilities exist, the Cook Ice Shelf is relatively dynamically stable under current conditions.

Despite this low vulnerability, ocean-driven basal melts were highly uncertain, highlighting the need for targeted observational field campaigns. Improved monitoring of the Cook Ice Shelf is important, given its role as the fundamental outlet glacier of one of the largest East Antarctic basin; the Wilkes Subglacial Basin (Lee et al., 2023). A weighted decision-making framework was developed for ApRES deployment, combining basal melt rates, crevassing, buttressing, and surface mass balance. This framework identifies a southern-central region near the grounding line ideal for instrument deployment. These data could be used to reduce uncertainties in melt parameterisations, and improve future ice sheet modelling. By linking this framework with fieldwork campaigns such as Antarctica InSync, this work supports a melt-targeted approach for monitoring the future vulnerability of the Cook Ice Shelf to climate change.

Chapter 7 – Open Data

All data and code used to calculate the above quantities and generate all figures are available at https://github.com/jeremaiahMON/COOK_ICE_SHELF_HONOURS and <https://zenodo.org/records/17100289>.

Chapter 8 – References

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Chapter 9 - Appendices

Appendix A – Ice Extent Area and Outflux

Table 3: The annual ice extent area (km²) and ice flow outflux (Gt/yr) of the Cook Ice Shelf. Note that some years of outflux data is absent due to insufficient velocity input data. The *Area* column is relevant to crevassing/calving, and the *Outflux* column is relevant to flow/buttrressing.

Year	Ice Extent Area (km ²)	Outflux (Gt/yr)
2006	2075.4549	0.34962
2007	2106.5455	-
2008	2156.8576	0.48226
2009	2172.1505	-1.0319
2010	2190.2559	-1.3186
2011	2229.174	-0.99754
2012	2262.3015	-
2013	2307.0813	-
2014	2341.7203	-0.91079
2015	2387.2583	-1.6868
2016	2438.845	-1.5131
2017	2477.9461	-0.942
2018	2513.4284	-0.72635

Appendix B – Literature Review

Aspects of Ice Shelf Vulnerability to Ocean-Driven Melt: A Review of Relevant Processes Affecting Ice Shelf Stability

Introduction

Anthropogenic global warming has prompted climate changes which negatively impact the Earth's cryosphere, accelerating melt and sea level rise. The evolution of ice shelves - fringing 75% of Antarctica's coastline (Rignot et al., 2013) - is central to this process, as they regulate ice flow into the ocean through buttressing, the resistive stabilising forces which ice shelves exert upon ice sheets, reducing the velocity of ice outflow (Greene et al., 2022) (figure 1). Previous studies emphasise that most Antarctic ice shelves are critical to the stability of ice flow, with Antarctic ice loss stemming from the basal melting and calving of ice shelves (Fürst et al., 2016). Both of these processes weaken buttressing and increase the ice flow velocity of upstream grounded ice (Reese et al., 2018). Basal melt of ice shelves is the dominant driver of

mass loss in regions such as the Amundsen Sea sector of West Antarctica, and is expected to increase around Antarctica over the coming decades to centuries. However, basal melt is critically undersampled and poorly constrained across Antarctica (Schodlok et al., 2016). Given its importance in understanding and modelling the past and future impacts of climate change on ice sheets (Song et al., 2025), there is a clear need for more wide-spread and accurate measurements of basal melt rates (McCormack et al., 2024). Losses in buttressing can also be caused by calving events through the propagation of crevasses (Greene et al., 2022). However, crevasse fields are not mapped in detail across Antarctica (Zhao et al., 2022), impacting the accuracy of predicting large-scale calving events. Therefore, there is a critical need to better understand these processes of melting, crevassing, and buttressing, and how they impact the flow of upstream ice, allowing for more accurate predictions of future global sea level rise.

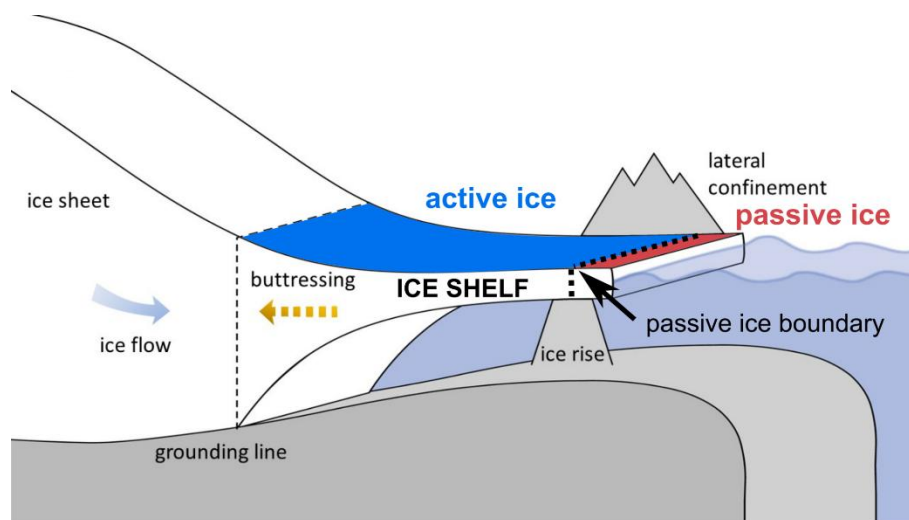


Figure 1: The grounded ice sheet transitions to a floating ice shelf at the grounding line - the point where the ice detaches from the bed and begins to float in the ocean. Buttressing forces that stabilise the upstream grounded ice sheet are generated due to the interaction between the ice shelf and topography, or lateral confinement in embayed ice shelves (modified from (Reese, 2019)).

Given the challenges of accurately measuring ice shelf basal melt rates, evaluating the strengths and limitations of existing observational approaches highlights the value of using an Autonomous phase-sensitive Radio Echo Sounder (ApRES) for high-resolution, in-situ measurements. Melt rates can be estimated using a range of observational techniques, each with advantages and disadvantages. Satellite remote sensing provides broad, but coarse temporal and

spatial resolution coverage. This instrument infers melt indirectly by detecting changes in ice shelf surface elevation through radar altimetry (Giles et al., 2008), but relies on steady-state conditions and hydrostatic equilibrium assumptions that introduce uncertainties (Nicholls et al., 2015). Oceanographic instruments from vehicle or ship-based instruments can provide estimates relating to ocean variables (e.g. temperature, salinity), but are limited to seasonality and logistical constraints (Smith et al., 2019). In contrast, ApRES is a remote sensing radar instrument which measures changes in vertical strain rates and basal melt rates (Nicholls et al., 2015). It is a ground-based sensor which has millimetre precision when measuring the depths of internal layers within an ice shelf (Lok et al., 2015). ApRES provides high temporal and spatial resolution estimates, and allows data to be characterised into ocean-driven basal melting and strain-induced thinning - a distinction that cannot be identified with satellite or oceanographic datasets alone. Through this, ApRES remains the most effective instrument for identifying local melt processes and validating in-situ measurements.

The overarching aim of this thesis is to assess the present-day vulnerability of the Cook Ice Shelf to climate change, through reduced buttressing in response to ocean-driven melt and crevassing. This is addressed through considering three core research questions: what is the magnitude and mode of ocean-driven basal melting, what regions are vulnerable to crevasse formation and subsequent ice calving events, and how has the flow regime evolved in response to ocean-driven melting and crevassing over recent decades and what do these changes reveal about the buttressing capacity of an ice shelf? This thesis focuses on the Cook Ice Shelf, which buttresses the Wilkes Subglacial Basin (WSB) and is likely vulnerable to rapid ice loss under future warming, and critically reviews past research on the ice shelf, ApRES, and associated melt processes, crevassing, and buttressing (Key Topics). Next, an overview of a decision-making process will be explained (Discussion), followed by a summarisation of the literature review (Conclusion).

Key Topics - The Cook Ice Shelf

The Cook Ice Shelf buttresses against the Cook Glacier in East Antarctica (figure 2). The upstream Cook Glacier is fed by one of the largest East Antarctic subglacial basins: the WSB, which spans 840,000 km² in area (Lee et al., 2023). The WSB has the potential to increase global sea levels by 3.02 m (Rignot et al., 2019). The grounding line of the Cook Ice Shelf is entirely below sea level and reaches depths around -1,400 m (Constantino and Tinto, 2023). This region is vulnerable to marine ice sheet instability due to the retrograde inland-sloping bed. This can lead to a positive feedback loop of increased ice outflow and sea level rise contribution, due to the retreat of the grounding line into deeper waters (Miles et al., 2018).

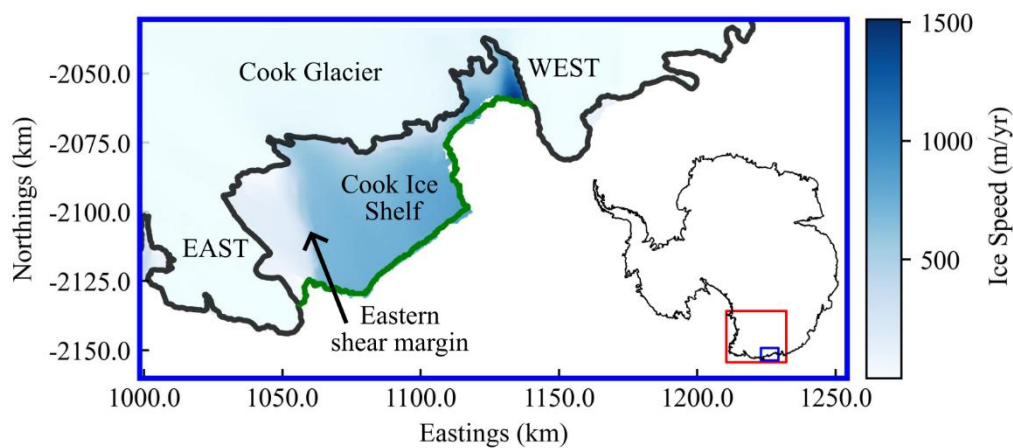


Figure 2: The spatial context of the WSB (red box) and the Cook Ice Shelf (blue box) in Antarctica. The ice velocity of the Cook Ice Shelf is presented (Rignot et al., 2017). The 2003 grounding line is plotted as a solid black line (Bindshadler and Choi, 2011) and the 2006 ice front is plotted as a solid green line (Greene et al., 2024).

The WSB is a region of potential dynamic instability under future climate warming, as evidence from paleoclimate records suggests that it experienced significant retreat and ice loss during past warm periods. The WSB saw an extensive retreat 330,000 years ago, and a smaller limited retreat 125,000 years ago. These events were linked to warmer interglacial Southern Ocean temperatures and higher global sea levels (Crotti et al., 2022). This is a cycle which could result in the future collapse of the WSB again.

Recent observations show substantial dynamic change across the Cook Ice Shelf, with western flow velocities increasing up to 22% between 2017 and 2022, indicating mass balance changes and the weakening of ice shelf stability mechanisms (Lee et al., 2023). This western region's vulnerability is evident from a near-complete collapse between 1973 and 1989, corresponding to the doubling of ice velocities (Miles et al., 2018). Historically, ice loss and instability of the WSB has been connected to grounding line retreat due to ice-ocean interaction melt (Mengel and Levermann, 2014), and increased ocean-driven melt from rising sea surface temperatures (Iizuka et al., 2023). Despite this, the drivers and spatial variability of melt, crevassing, and buttressing remain poorly constrained. These uncertainties limit understanding of how the Cook Ice Shelf - WSB system will respond to future oceanic and atmospheric forcings, motivating the need for further targeted investigation into basal melting processes and ice shelf stability.

Key Topics - ApRES

ApRES is a ground based radar instrument which measures vertical strain thinning and basal melt rates. This radar records a time series of amplitude and phase versus depth, enabling the tracking of internal layers through time (Cook et al., 2022). By analysing repeated nadir observations, changes in the depths of internal and basal reflectors can be quantified to derive basal melt rates and vertical strain thinning (Vaňková et al., 2020). The ApRES can be configured in different modes depending on the observational goals: a monostatic point-ranging configuration which provides a one-dimensional measurement at a single location, while an orthogonal multiple-input multiple-output array allows for two-dimensional data through ice-imaging (Lok et al., 2015). However, ApRES measurements and their accuracy can be compromised by surface roughness or crevassing, which introduce noisy basal reflections (McCormack et al., 2024). Additionally, to accurately derive basal melt rates and vertical strain

thinning, surface accumulation between site visits must be corrected for by accounting for surface mass balance (Nicholls et al., 2015).

Key Topics - Ocean-driven ice shelf melt rates and ApRES

Basal melting - the ocean-driven melting at the base of floating ice shelves - plays a central role in controlling ice thickness, flow dynamics, and grounding line migration (Vaňková et al., 2021). Basal melt rates are driven by warming from the ocean, predominantly through three main ‘modes of melting’. Each mode represents a different water parcel type, identified according to specific temperature, salinity, and neutral density definitions. Mode 1 involves colder, saline Dense Shelf Water (DSW) formed in coastal polynyas that sinks to deeper depths, causing basal melt near grounding lines (McCormack et al., 2024, Adusumilli et al., 2020). Mode 2 is the most common and dominant mode of melting. This involves the polewards transportation of Circumpolar Deep Water (CDW) by changing wind stress patterns due to climate change, where CDW is moved and mixed across the continental shelf to form modified Circumpolar Deep Water (mCDW). This parcel of water is warmer than DSW, resulting in more intense basal melt rates (McCormack et al., 2024, Adusumilli et al., 2020). Mode 3 refers to melt by Antarctic Surface Waters (ASW), driving seasonal melting near ice shelf fronts at shallower depths (McCormack et al., 2024, Adusumilli et al., 2020). While almost all Antarctic ice shelves experience basal melting, thinning only occurs when basal melt exceeds steady-state mass balance, leading to reduced buttressing, dynamic ice loss, and rising sea levels (Adusumilli et al., 2020).

Modelled estimates of basal melt rates at the Cook Ice Shelf is relatively low across the shelf, but remains one of the greatest uncertainties due to limited in-situ observations. Existing studies indicate heterogeneous melt-induced thinning patterns, with higher basal melt rates identified

in the far west and along the grounding line of the central-western body of the shelf (Paolo et al., 2024). The overall basal melt rates of the Cook Ice Shelf are relatively low, with shelf averages under 2 m/yr (Jordan et al., 2022). These modelled estimates rely on height to thickness inversions and solving for mass balance equations (Paolo et al., 2024), and assumptions of steady state and hydrostatic equilibrium, which introduce uncertainties. There is a lack of in-situ fieldwork data to verify the accuracy of modelled basal melt rates. The presence of mCDW is suspected (Constantino and Tinto, 2023), however this has yet to be spatially constrained and linked to basal melt rates. The last oceanographic in-situ data dates back to 1957-58, where no mCDW was observed at the time. There has been a lack of ship-based conductivity-temperature-depth surveys since then (Constantino and Tinto, 2023), creating uncertainties in melt rates due to this sparsity of observations. Therefore, melt remains one of the largest uncertainties in this Cook Ice Shelf - WSB basin, and needs to be verified by in-situ data.

ApRES can be utilised to understand the characteristics of melt by providing continuous measurements of melt rates through time. These measurements not only quantify the magnitude and spatial variability of melting, but also serve as indirect indicators of the oceanographic processes driving it. Because basal melt rates reflect the exchange of heat and (loss of) mass at the ice-ocean interface, variations detected by ApRES can reveal changes in ocean circulation, and the influence of different water masses within the sub-ice-shelf cavity (Nicholls et al., 2015). Moreover, by using ApRES to identify changes in ice layer depths (Oraschewski 2024), the thinning of ice and deformation can be observed, indicating the vulnerability of the ice to a melt regime. It can also be used as a proxy for dynamic thinning, which can lead to reduced buttressing to upstream ice flow, increase discharge of grounded ice, and therefore a rise in sea levels (Gudmundsson et al., 2019). By providing high resolution timeseries data of basal melt

rates and vertical strain changes, ApRES can further assist in identifying the subsequent impacts on the flow regime of the ice.

Key Topics - Crevassing and calving

Crevasses are features of ice shelves that play a key role in controlling shelf vulnerability and the calving of ice into the ocean. Crevasses are brittle fractures that form from the stresses within an ice body (Colgan et al., 2016), typically in regions of extension and higher strain rates, when these rates exceed the fracture toughness of the ice (Thompson et al., 2020). On ice shelves, two main types of crevasses are recognised; surface and basal. Surface crevasses occur near the top of the ice shelf, while basal crevasses initiate at the ice-ocean interface (Luckman et al., 2012). The formation of basal crevasses is caused by tensile stresses and high basal water pressures. When small basal cracks exceed a threshold size, they can rapidly propagate and extend upwards (Luckman et al., 2012). These fractures play a crucial role in calving, the process by which large sections of ice detach from the shelf front (Benn and Åström, 2018). Calving represents a natural cycle in ice-shelf evolution that may become more frequent under a warming climate.

At the Cook Ice Shelf, observations between 1947 and 2017 show minimal crevassing in the eastern region, in contrast to extensive crevassing across the western section (Miles et al., 2018). Furthermore, the far western region of the Cook Ice Shelf experienced a near-complete calving event loss between 1973 and 1989, corresponding to a doubling in the ice velocities of this region (Miles et al., 2018). Despite this dynamic loss in the western region, the eastern Cook Ice Shelf has observed no major calving event since 1973, with the ice front instead steadily advancing over time (Miles et al., 2018). The observed surface crevassing and calving is limited to satellite imagery observations, which leave surface crevasse identification up to individual

user visual tracing. There is also the absence of literature material relating to assessing the probability/vulnerability of regions of the Cook Ice Shelf to crevasse formation.

ApRES can detect areas of high stress and deformation within ice shelves (through strain thinning), enabling the identification of regions vulnerable to crevasse formation and future calving events. The ability to observe internal layer changes and deformation also allows for the detection of subsurface structures (anomalies) that are not visible from satellite imagery, but could still act as predictors for crevassing initiation. Additionally, repeated ApRES measurements provide a trend of strain evolution over time, which can help distinguish the propagation of small fractures into potential calving events. However, a significant limitation of ApRES deployment use in crevassed regions is the consequent scattering of radar signals, resulting in noisy data quality and difficulty in signal interpretation.

Key Topics - Ice shelf flow response of buttressing

Buttressing ice shelves restrain the outflow of upstream grounded ice from ice sheets. A reduction in this buttressing capacity accelerates ice outflow, contributing to sea level rise (Fürst et al., 2016). Buttressing also stabilises the grounding line by reducing ice flux and mass loss (Haseloff and Sergienko, 2018). However, the buttressing capacity of an ice shelf can be heterogenous. *Active ice* exerts resistive forces on upstream ice, while *passive ice* does not, with the passive ice boundary separating these regions. Buttressing can be quantified using the *buttressing number*, where values ≤ 0 indicate passive ice, > 0 represents active ice, and ≥ 1 signifies complete buttressing.

Ice shelf buttressing critically controls grounded ice flow, but melt-induced thinning and calving can weaken this restraint, increasing ice discharge and sea level rise (Gudmundsson et al., 2019). High melt rates correspond to increases in stress and strain levels (Watkins et al.,

2024), causing extensional thinning that weakens shelf structure and reduces buttressing. Crevassing and calving further reduce buttressing. A loss of passive ice has little effect, whereas calving of active, highly buttressed ice accelerates upstream ice discharge (Fürst et al., 2016).

The near-complete loss of the western Cook Ice Shelf between 1973 and 1989 corresponds to a 100% increase in ice flow velocity of this region (Miles et al., 2018). This calving reduced buttressing, doubled upstream ice discharge, and caused dynamic thinning inland, as ice mass is redistributed towards the ice front as extensional stress increased. This is reflected in satellite imagery observations, with inland thinning upstream of the western Cook Ice Shelf identified between 1992 and 2010 (Miles et al., 2018), reinforcing the link between calving, reduced buttressing, accelerated ice flow, and increased sea level rise.

Buttressing across the Cook Ice Shelf varies spatially, and is strongest along the eastern shear margin, the grounding line, and the south-western sector, while the northern regions portrays weaker buttressing that diminishes towards the ice front (Durand et al., 2016). Existing datasets of a 1996 - 2009 averaged buttressing number calculation and a single mapped passive ice boundary (Fürst et al., 2016) lack the spatial and temporal resolution to capture short-term changes in the shelf's flow regime. Despite this, evidence of dynamic variability is observed. The eastern body of the Cook Ice Shelf from 2006 - 2007 saw a 12% increase in ice velocity (compared to its 2001 - 2016 average) - likely due to basal melting - and a consequential subglacial flood event (Miles et al., 2018). This short-lived increase in velocity correlated to a subglacial lake drainage discharge of 5.2 km³ of water originating ~ 450 km upstream of the eastern Cook Ice Shelf.

ApRES can estimate the buttressing capacity of the Cook Ice Shelf by measuring strain thinning, linking ice column thickness reduction to vertical velocity profiles (Vaňková et al., 2021), and inferring ice flow velocities. Moreover, increased strain thinning can indicate

increased vulnerability to dynamic ice flow. By deploying ApRES instruments in regions of active buttressing, the vertical strain thinning for regions that directly impact the acceleration and discharge of upstream grounded ice can be measured. Therefore, the implementation of ApRES provides direct monitoring of variations in shelf thinning and internal deformation, and how these translate into changes in buttressing strength, offering a dynamic understanding of ice shelf and ice sheet interactions at the Cook Ice Shelf.

Key Topics - Decision-making framework

The deployment of ApRES instruments across the Cook Ice Shelf must be guided by a clear decision-making framework that aligns with the three research questions, in order to assess the present-day vulnerability of the Cook Ice Shelf to climate change, through reduced buttressing in response to ocean-driven melt and crevassing.

To address the first aim of targeting the magnitude and mode of ocean-driven basal melting, ApRES sites can be selected using modelled basal melt rate maps to identify regions of high melt rates. Areas overlying sub-shelf cavities may be of interest, as the upwelling of ocean water can drive basal melting. Although regions near the grounding line often exhibit high basal melt rates and exert strong control on ice shelf dynamics (Lambert et al., 2022), they may not always be suitable for ApRES deployment due to increased horizontal shear and signal distortion.

For the second aim focusing on vulnerability to crevassing and calving, ApRES deployment should avoid regions with heavy crevassing as these features can create off-nadir radar reflections that complicate signal interpretation and increase uncertainties (Vaňková et al., 2021). Strain rate maps can serve as useful proxies to locate areas prone to crevasse formation, while digital elevation models can help exclude sites with complex basal topography,

Finally, to address the third aim of understanding how the Cook Ice Shelf flow regime and buttressing have evolved, ApRES deployment can be inferred from satellite velocity measurements to monitor strain thinning and vertical velocity profiles over time. When selecting deployment sites, it is important to consider the buttressing capacity of each region, as this determines how directly ice flow responds to melt-induced thinning and crevasse formation.

Overall, effective ApRES deployment is influenced by many factors, and depends on balancing data quality, instrument safety, and scientific return, to achieve a more holistic understanding of the Cook Ice Shelf's dynamic vulnerability to climate change.

Conclusion

Ultimately, understanding the interconnection of melt processes, flow regimes, and crevassing is critical to assessing the stability and vulnerability of the Cook Ice Shelf to ocean-driven melt. The melt processes relate to the mode and magnitude of melt, and the flow regime refers to the subsequent impacts from this melting and the changes in buttressing. Crevassing processes can also lead to mass-ice-loss calving events. These three themes can be analysed by ApRES through measuring the basal melt rate and strain thinning. Combining these processes with satellite datasets support the development of a decision-making process for relevant ApRES deployment.

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